Highlights

Position-scheduled linear parameter-varying control of thin-walled plate milling: certified stability lobes, saturation-aware certificates, and the limits of regenerative-targeted delayed feedback

Author One, Author Two

- Tool position and material removal are scheduled rather than treated as uncertainty
- Certified tuning shows the optimal delayed-feedback gain is zero under scheduling
- Worst-position stable depth of cut 2.99 mm versus 0.70 mm for the best frozen design
- Solver-free saturation certificate matches the measured clip onset within 2 percent
- Saturation certification overturns the linear ranking and predicts saturation islands

Position-scheduled linear parameter-varying control of thin-walled plate milling: certified stability lobes, saturation-aware certificates, and the limits of regenerative-targeted delayed feedback

Author One, Author Two

Abstract

In thin-walled workpiece milling the structural dynamics seen by the cutting force vary strongly and predictably with the tool position along the feed path and with progressive material removal, yet the prevailing active chatter-control practice wraps this known variation into norm-bounded uncertainty and pays for it with conservatism. This paper develops a position-scheduled linear parameter-varying (LPV) H_∞ output-feedback controller for a piezoelectric-patch-actuated thin-walled plate, gain-scheduled on the tool position and the material-removal state—both known in real time from the NC program and CAM model—while norm-bounded uncertainty is retained only for what is genuinely uncertain: truncated-mode spillover, cutting-coefficient dispersion, and modal tolerances. A spindle-synchronized delayed feedback term acting on the observer-reconstructed milling-point displacement targets the regenerative force directly; its gain is tuned offline by maximizing the closed-loop critical depth of cut computed by semi-discretization, with a smallest-gain preference so the term can never degrade the certified loop. The tuning delivers an unreported finding: under the active scheduled loop the certified optimal delayed gain is zero at every scheduling point, while the same delayed action applied alone destabilizes the process at other spindle speeds. Stability of the scheduled, time-periodic, delayed closed loop—including controller, actuator roll-off, and implementation-latency dynamics—is certified by semi-discretization stability lobes along the tool path, complemented by a small-gain spillover margin and an exact sampled-loop spectral-radius test. On a validated Mindlin finite-element model of a benchmark aluminium plate (natural-frequency errors of 1.5–3.8% against published measurements, FRF-level agreement with the rig's measured receptance and piezoelectric transfer function, and reproduction of its measured open-loop stability pattern—including its 5500 rpm anomaly—once the published force calibration is applied), the scheduled controller raises the worst-position critical depth of cut from 0.70 mm for the strongest non-scheduled H_∞ design of the same family to 2.99 mm—a de-conservatization factor of 4.2, and thirteen times the open-loop limit—and at the end-of-pass condition where the frozen design is weakest it sustains nearly three times that design's local stability limit with 42% of the vibration amplitude at half the control voltage. Because that voltage budget is finite, the same deployed loop is then certified against actuator saturation: a sampled-data period lifting gives, solver-free, the largest surface-defect perturbation for which the amplifier provably never clips, matching an independent nonlinear simulator's measured clip onset to 0.01–1.8%. Swept into a certified permissible depth-of-cut envelope, the certificate reorders the two designs: the scheduled controller's linear worst-position advantage (4.2-fold above; 3.2-fold in the sampled-data lift used here) inverts, in *guaranteed saturation-free depth*, to 0.79-fold at a loose 20 μm tolerance, even as its realized vibration and voltage superiority persists—and two causal saturation islands, reproduced also on a separate published rig, turn actuator saturation from a hidden failure mode into a certified planning constraint.

Keywords: Milling chatter, thin-walled workpiece, active vibration control, linear parameter-varying control, delayed feedback, semi-discretization, actuator saturation, maximal output admissible set, saturation island

---

1. Introduction

Regenerative chatter limits the productivity of milling operations on thin-walled parts more severely than on any other workpiece class: the low modal mass and light damping of a thin wall depress the critical axial depth of cut to a fraction of a millimetre, and the stability boundary itself shifts as the tool traverses the part [1, 2]. Recent reviews of thin-wall machining dynamics agree that the in-process variation of the workpiece dynamics—driven by the tool position along the feed path and by progressive material removal—is the core unresolved difficulty for active chatter suppression, and they explicitly call for adaptive or scheduled control solutions [3, 4, 5]. That the variation is physically well characterized and predictable is established: stability maps resolved along the tool path are standard practice in open-loop process planning [6], surrogate models of the varying workpiece dynamics are available for chatter prediction [7], and linear parameter-varying (LPV) descriptions of thin-wall milling dynamics over tool position and material removal have been published [8].

On the actuation side, piezoelectric elements bonded to or acting on the flexible workpiece are the established hardware route [9, 10, 11, 12]. The dominant control paradigm built on this hardware is robust, worst-case design: the position-dependent modal participation, the intermittent cutting-force coefficients, and the modal drift caused by material removal are lumped into norm-bounded uncertainty around a single nominal plant, or handled by a single design at the worst operating point. Representative examples include μ-synthesis and H_∞ designs for milling and turning [13, 14, 11, 15, 16, 17, 18, 19], optimal control of the regenerative process dynamics through feed drives [20, 21], and multi-delay robust formulations for variable-pitch cutters [22]. The state of the art for the plant class considered here is the work of Du et al. [13], who combine a μ-synthesis controller with a delayed PD term on a piezo-patch-actuated thin-walled plate and treat *all* in-process variation as norm-bounded uncertainty. This paradigm is deliberately and provably conservative: robustness against variation that is in fact known is paid for in attainable depth of cut and in control voltage.

Scheduled, adaptive, and learning alternatives exist but are sparse. Wang et al. [12] vary PD gains with tool position on exactly the thin-walled-plate/piezo-patch plant class, but heuristically, following the first mode shape, with no synthesis machinery, no robustness or spillover treatment, and no stability certificate. A recent spindle-speed-mapped framework [23] optimizes saturation-aware delayed displacement-difference feedback gains per spindle speed and stores them as a gain map, with experimental validation; the scheduling variable is the spindle speed, not the tool-path position or the material-removal state, and no LPV synthesis or scheduling-consistent stability machinery is invoked. Brand et al. [24] report a robust LPV H_∞ design for an active tool holder in internal turning, scheduled on a quasi-static setup parameter (tool overhang); turning presents a single constant delay and an LTI plant at each operating point, and their controller carries neither a scheduled delayed-feedback term nor a certificate for a time-periodic delayed closed loop. Adaptive schemes identify and re-tune online [25, 26], reacting to variation rather than exploiting a priori NC-program knowledge, and provide no closed-loop stability certificate for the delayed periodic system; learning-based controllers [27] explicitly forgo model-based guarantees.

LPV gain scheduling itself is mature in adjacent machine-tool components—feed drives [28], ball screws [29]—and in smart structures [30], with a well-developed theory [31, 32, 33, 34] including synthesis conditions for delayed LPV systems [35]; in milling, the phrase “time-delay LPV control” has so far been attached only to a lumped two-degree-of-freedom model scheduled on cutter angle [36].

A second, largely separate lineage feeds this work: delayed feedback that targets the regenerative mechanism itself. The delayed-resonator concept [37, 38, 39], delayed feedback distorting the regenerative effect in turning [40], spindle-synchronized displacement-difference feedback in milling [41, 42], delayed output feedback with stability-lobe shaping [16], and optimal delayed state feedback for a thin-walled part through an active fixture [43] all demonstrate that a delay term locked to the tooth-passing period is a powerful and hardware-free lever. In every published instance, however, the delayed gain is fixed or robustly designed—or, at most, mapped over spindle speed [23]—never scheduled on the tool position or the material-removal state.

Finally, computing stability lobes *with the controller in the loop* is established verification practice: closed-loop stability lobe diagrams [44], lobes including actuator, sensor, and filter dynamics—whose omission is shown to falsify the predicted boundary [45]—and experimentally validated model-based closed-loop charts [46, 47]. The underlying semi-discretization machinery [48, 49, 50] demonstrably handles periodic coefficients and modulated gains, and robustness extensions for time-periodic delayed systems exist [51]. No published work applies this machinery to certify a *parameter-varying, scheduled* delayed closed loop along a tool path.

The gap addressed here follows from a simple methodological observation: the dominant “uncertainty” in thin-wall milling is not uncertain at all. The tool position along the feed path is known in real time from the NC program and the drive encoders, and the material-removal state is a known monotone function of that position through the CAM model. A controller that schedules on these variables does not need to buy robustness against them. The present paper therefore replaces the prevailing “known variation as uncertainty” treatment with “known variation as scheduling, residual variation as uncertainty.” It then confronts a second idealization that binds precisely where the first pays off—at aggressive depth of cut: the design and its stability certificates assume a linear actuator, whereas the piezoelectric amplifier is bounded and saturates under demand, a regime in which recent measurements report “saturation islands” of chatter erupting below the linearly predicted boundary once the actuator clips [52]. The deployed loop is accordingly certified against its own amplifier bound, and the two themes—scheduling and saturation—together make four contributions.

1. **C1 — Known variation as scheduling, residual variation as uncertainty.** A grid-based LPV H_∞ output-feedback controller for piezo-patch-actuated thin-walled workpiece milling is gain-scheduled on the tool position along the feed path and on the material-removal state—both known in real time from the NC program / CAM model—while a residual additive uncertainty block covers scheduling-map error and truncated-mode spillover. This de-conservatizes the robust baseline [13] that wraps the same known variation into uncertainty; the conservatism gap is quantified in stable depth of cut and in γ-level as a function of position.
2. **C2 — A certified assessment of regenerative-targeted delayed feedback under scheduling.** The spindle-synchronized delayed displacement-difference gain—previously fixed [41, 43] or mapped over spindle speed [23]—is embedded in the scheduled architecture and tuned offline by maximizing the *closed-loop* critical depth of cut computed by semi-discretization of the full time-periodic delayed closed loop, controller dynamics included, with a preference for the smallest gain among ties so the term can never degrade the certified loop. The certified tuning delivers a finding the delayed-feedback literature has not reported: once the scheduled H_∞ loop is active, the optimal delayed gain is *zero* at every scheduling point of this benchmark—and the same low-order delayed action applied alone destabilizes the process at other spindle speeds. The protocol, not a claimed increment, is the contribution.
3. **C3 — Certification of the scheduled, time-periodic, delayed closed loop.** Stability of the parameter-varying, time-periodic, delay-differential closed loop—with full controller, actuator, and filter dynamics—is certified by semi-discretization stability lobes along the tool path under a bounded scheduling rate. This certificate, not the grid LPV synthesis itself, is the formal stability guarantee, extending closed-loop stability-lobe practice [44, 45] to scheduled controllers.
4. **C4 — Saturation-aware certification of the deployed loop.** The same deployed controllers are certified against the amplifier deadzone by a sampled-data period lifting whose maximal saturation-free admissible set gives, in closed form and without a semidefinite solver, the

largest surface-defect perturbation for which the loop provably never clips. The certificate is implementation-exact—control rate, computation delay, and amplifier bound—and matches an independent nonlinear simulator's measured clip onset to 0.01–1.8%; its sweep is a certified permissible depth-of-cut envelope that is shown to *overturn* the linear controller ranking as the declared surface-defect tolerance grows. The certificate predicts and explains saturation islands, and reproduces the published island measurements of a separate rig [52] on their own parameters. This extends the maximal-output-admissible-set construction [53] and the saturated-delay certificate lineage [54, 55, 56] to milling for the first time, on the true time-periodic delayed dynamics rather than the averaged model of [57].

Every individual ingredient—LPV scheduling, H_∞ synthesis, delayed feedback, closed-loop stability lobes—exists separately in the literature, and the contribution claimed here is their integration under the scheduling paradigm—and, as a fourth contribution, a saturation-aware certificate of the deployed loop—with precise differentiation from the closest prior works. Relative to Brand et al. [24], the present plant is a time-periodic milling loop with a regenerative delay, scheduled on a real-time trajectory rather than a quasi-static setup parameter, with a semi-discretization certificate and a certified assessment of the delayed-feedback add-on. Relative to Wang et al. [12], the position-varying-gain idea is generalized from a heuristic single-mode PD law to a certified multivariable synthesis with explicit spillover safety. Relative to the spindle-speed-mapped framework [23], the scheduling variables are the tool-path position and material-removal state and the gain map is replaced by grid LPV synthesis with well-posed interpolation. Relative to Du et al. [13], the same plant and actuator are retained and the entire contribution is the quantified de-conservatization. Relative to Maslo et al. [8], the LPV description of thin-wall milling dynamics is taken from modeling into certified active control synthesis. Finally, relative to the saturation-island measurements of Ozsoy et al. [52] and the constraint-avoiding, spindle-speed-mapped gain maps of [23], the present work supplies the missing regional guarantee: a computable, maximized saturation-free set for a *given* controller behind a *given* amplifier on the unaveraged time-periodic delayed dynamics.

The remainder of the paper is organized as follows. Section 2 develops the electromechanical model and its validation. Section 3 presents the controller synthesis, the scheduled delayed feedback term, the deployment certificates,

Table 1: Model and process parameters (following the experimental setup of [13]).

Quantity	Symbol	Value	Unit
Plate length / height / thickness	l_p, h_p, b_p	100/80/4	mm
Plate modulus / Poisson / density	E_p, ν_p, ρ_p	69 / 0.33 / 2830	GPa, –, kg/m^3
Modal damping ratios (modes 1–5)	ζ_i	0.31/0.17/0.27/0.56/0.35	%
Patch dimensions	–	$60 \times 20 \times 0.7$	mm
Patch modulus / Poisson	E_{pe}, ν_{pe}	63 / 0.35	GPa, –
Patch strain constant	d_{31}	-175	pm/V
Amplifier voltage limit	$V_{\max}$	± 150	V
Sensor noise (RMS)	–	0.2	μm
Tool: flutes / diameter / helix / rake	N_t, D, –, –	3 / 10 / 35 / 15	–, mm, deg, deg
Cutting coefficients	k_t, k_n, μ_c	925 / 0.26 / 0.2	MPa, –, –
Radial depth / feed per tooth	a_e, f_t	0.1 / 0.02	mm, mm
Reference spindle speed / axial depth	Ω, a_p	4900 / 0.3	rpm, mm

and the baselines. Section 4 details the semi-discretization analysis of the scheduled delayed closed loop. Section 5 reports the linear closed-loop results, Section 6 develops and applies the saturation-aware certification, Section 7 discusses limitations, and Section 8 concludes; Appendix A reports the external reproduction of the published saturation islands.

2. Electromechanical model of thin-walled plate milling

2.1. Plant configuration and physical data

The benchmark plant reproduces the experimental setup of the reference baseline [13] so that simulation results are directly comparable with published measurements: a cantilever AL6061 plate of length $l_p = 100$ mm (feed direction x), height $h_p = 80$ mm (cantilever span z), and thickness $b_p = 4$ mm, clamped along its bottom edge $z = 0$ and milled in down-milling along its free top edge $z = h_p$ with feed in $+x$. A piezoelectric patch (QDA60-20-0.7, $60 \times 20 \times 0.7$ mm) is surface-bonded near the clamped lower-left corner and driven by a ± 150 V amplifier; an eddy-current displacement probe at the fixed point $(x_s, z_s) = (95, 75)$ mm provides the single measurement. The physical and process data are collected in Table 1.

2.2. Finite-element plate model and piezoelectric actuation

The plate is discretized with 4-node Reissner–Mindlin quadrilateral elements with nodal degrees of freedom (w, θ_x, θ_z) and selective reduced integration of the transverse-shear terms to avoid locking. The mesh is 40×32 elements (element size 2.5 mm, aligning the patch edges with element boundaries). Assembly yields

$$\mathbf{M}\,\ddot{\mathbf{q}}(t) + \mathbf{C}\,\dot{\mathbf{q}}(t) + \mathbf{K}\,\mathbf{q}(t) = \boldsymbol{\Theta}\,u(t) + \mathbf{e}_w(x_T)\,F_y(t), \tag{1}$$

where u is the patch voltage, F_y the transverse milling force, and $\mathbf{e}_w(x_T)$ the nodal interpolation vector of the transverse deflection at the milling point (x_T, h_p). Damping is not assembled: it is imposed modally with the measured ratios of Table 1, since Rayleigh damping cannot match five arbitrary ratios and modal damping is exact by construction in the reduced model.

The surface-bonded patch is modeled by the standard induced-moment mechanism for a thin bonded actuator: the voltage induces the isotropic in-plane strain $\Lambda = d_{31}u/h_{\text{pe}}$ acting at the offset $z_{\text{off}} = (b_p + h_{\text{pe}})/2$ from the plate mid-plane, producing the consistent nodal load

$$\mathbf{f}_{\text{pe}} = \boldsymbol{\Theta}\,u, \qquad \boldsymbol{\Theta} = \sum_e \int_{A_e \cap A_{\text{pe}}} \mathbf{B}_b^{\mathsf{T}}\,\mathbf{D}_{\text{pe}}\,\hat{\boldsymbol{\Lambda}}\;z_{\text{off}}\,\mathrm{d}A, \qquad \hat{\boldsymbol{\Lambda}} = [1,\,1,\,0]^{\mathsf{T}}\frac{d_{31}}{h_{\text{pe}}}, \tag{2}$$

with $\mathbf{B}_b$ the bending strain–displacement matrix and $\mathbf{D}_{\text{pe}}$ the plane-stress constitutive matrix of the patch. In the thin-patch limit, Eq. (2) reduces to the pin-force model used in the reference [13].

2.3. Modal reduction and parameter-varying structure

With the mass-normalized mode matrix $\Phi = [\varphi_1 \cdots \varphi_n]$ and $\mathbf{q} = \Phi\eta$, the reduced model reads

$$\ddot{\eta} + \mathrm{diag}(2\zeta_i\omega_i)\,\dot{\eta} + \mathrm{diag}(\omega_i^2)\,\eta = \Phi^{\mathsf{T}}\boldsymbol{\Theta}\,u + b_f(x_T)\,F_y, \qquad w_{\text{mill}} = b_f(x_T)^{\mathsf{T}}\eta, \qquad y_s = c_s^{\mathsf{T}}\eta + v, \tag{3}$$

where $b_f(x_T) = \Phi^{\mathsf{T}}\mathbf{e}_w(x_T)$ is the modal participation vector of the milling point—the sole entry point of the tool position into the dynamics—w_{mill} is the milling-point deflection (the chatter variable), and v is the sensor noise. Because the sensor location is fixed, the output map c_s is *constant*: scheduling enters only the dynamics, not the measurement, a structural advantage for scheduled observer-based control.

The design model retains the $n = 5$ measured modes of the reference rig. Truncation after mode 5 leaves a frequency gap of a factor 1.55 to mode 6 (4.04 $\to$ 6.28 kHz), which the actuator roll-off filter of Section 3.2 can separate; truncation after mode 3 would leave only a factor 1.23 to mode 4, which no realizable filter can separate (verified: three-mode designs exhibit catastrophic spillover on the full evaluation model). The evaluation model retains $n_{\text{full}} = 12$ modes (truncated band 6.3–12.2 kHz); modes beyond the measured five carry a damping ratio of 0.35%.

2.4. Regenerative milling force

The linearized single-frequency regenerative model is used for design, with plate flexibility dominant in the transverse direction (the tool compliance is negligible with respect to the plate, as in [13]). The classical linear-edge force model [2], integrated over the engaged helical flute segments of the three-flute tool in the down-milling engagement window $\phi \in [\pi - \phi_e, \pi]$ with $\phi_e = \arccos(1 - 2a_e/D)$, yields the transverse force

$$F_y(t) = \alpha_4(t)\,[w_{\text{mill}}(t-\tau) - w_{\text{mill}}(t)] - \alpha_3(t)\,f_t, \qquad \tau = \frac{60}{N_t\,\Omega}, \tag{4}$$

where $\alpha_4(t)$ [N/m] is the periodic regenerative (cutting-stiffness) coefficient, $\alpha_3(t)$ the static feed-ripple coefficient, and τ the tooth-passing period (4.08 ms at 4900 rpm). The coefficient is *signed*: for this down-milling engagement the directional factor makes $\alpha_4(t) \leq 0$, and the sign is carried consistently through synthesis, stability analysis, and simulation. The time-averaged value $\bar{\alpha}_4$ is used for design; the full periodic coefficient, together with the loss-of-contact (unilateral chip) nonlinearity, is used for evaluation. Substituting Eq. (4) into Eq. (3) gives the design-level delay-differential plant

$$\ddot{\eta} + \operatorname{diag}(2\zeta_i\omega_i)\,\dot{\eta} + \left[\operatorname{diag}(\omega_i^2) + \bar{\alpha}_4\,b_f b_f^{\mathsf{T}}\right]\eta - \bar{\alpha}_4\,b_f\,w_{\text{mill}}(t-\tau) = \Phi^{\mathsf{T}}\boldsymbol{\Theta}\,u - \bar{\alpha}_3 f_t\,b_f. \tag{5}$$

The cutting-stiffness dyad $\bar{\alpha}_4(a_p)\,b_f(x_T)\,b_f(x_T)^{\mathsf{T}}$ is rank one, position dependent, and proportional to the axial depth of cut: it is the LPV core of the problem.

2.5. Material removal and the scheduling vector

Milling the top edge at radial depth a_e per pass recedes the free edge: after machining up to position x_T, the plate height is $h_p - a_e$ for $x < x_T$. The finite-element model is rebuilt on the receded, remeshed geometry at a grid of removal states, and the modal sensitivities to removal are tabulated. For a single pass (0.1 mm off an 80 mm plate) the modal drift is small and is additionally covered by the retained parametric uncertainty; the removal scheduling matters over multi-pass sequences, which are simulated up to a total edge recession of 1 mm to exercise contribution C1 at full strength.

The scheduling vector is

$$\theta(t) = \big(x_T(t),\, \varrho(t)\big) \in \mathcal{P}, \tag{6}$$

with x_T the tool position, known in real time from the NC program and the drive encoders, and ϱ the material-removal state, known from the process plan (CAM model); $\mathcal{P}$ is a compact rectangle. At 4900 rpm the table feed is $f_t N_t \Omega/60 = 4.9$ mm/s, so one pass over the 100 mm plate takes 20.4 s, while the slowest structural period is 1.9 ms (mode 1 at 540 Hz): during one tooth-passing period the tool advances 20 μm (0.02% of the path), and the normalized scheduling rate $|\dot{\theta}| \approx 0.05\ \mathrm{s}^{-1}$ lies roughly four orders of magnitude below the slowest structural dynamics. The frozen-parameter (quasi-LPV, slowly varying) design regime [31, 32] is therefore quantitatively justified for this process; its residual gap to a global certificate is closed a posteriori in Section 4.

2.6. Model validation

Table 2 compares the natural frequencies of the finite-element model with the measured values reported for the reference rig and with the reference paper's own theoretical model [13]. The frequency errors of the present model are 1.5–3.8%, comparable to the 0.5–3.2% of the reference's model. Mesh convergence was verified: the 40×32 mesh reproduces the frequencies of a 60×48 mesh within 0.15%.

The validation extends beyond tabulated frequencies to the measured frequency-response curves themselves. The measured receptance and the piezo voltage-to-displacement transfer function of the reference rig (its Fig. 12) were digitized (89 and 76 points, 5 Hz–5 kHz; the point data ship with the code repository) and are overlaid on the model in Figure 1. Three levels of agreement result. First, the digitized resonance frequencies (536/1072/2779/3359/4123 Hz)

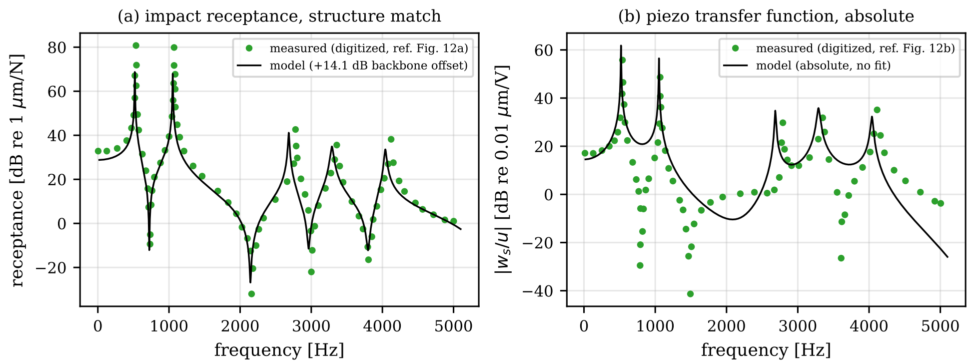

Figure 1: FRF-level validation against the measured curves of the reference rig (points: digitized from [13], Fig. 12): (a) impact receptance at the upper corner—model with a single fitted backbone gain offset (axis-reference ambiguity of the digitization, stated in the legend); (b) piezo voltage-to-displacement transfer function in absolute units with no fitted parameter.

independently confirm the tabulated measured values, and the model tracks them with the same 1.6–3.8% offsets as Table 2. Second, the *structure* of the receptance—including the antiresonance placements near 0.7, 2.2, 3.0, and 3.8 kHz, which probe the mode-shape ratios at the measurement point rather than the frequencies—is reproduced across the full band once a single constant gain offset is fitted (the digitized amplitude axis of the receptance plot carries a reference ambiguity, disclosed as such); a scale-free check on the same curve, the relative level of the first two resonance peaks, agrees to 0.1 dB (measured 0.9 dB, model 0.8 dB). Third, the actuation path is compared in *absolute* units with no fitted parameter: with handbook piezoelectric constants the model's low-frequency gain is 0.053 μm/V against the measured 0.072 μm/V (-2.6 dB, i.e. within 35% for the complete electromechanical chain), and the model backbone runs through the measured points across the band. Digitized resonance tips underestimate the true peaks—a 0.2–0.6% damping resonance is a few hertz wide, unresolvable from a printed plot—so peak amplitudes are compared as lower bounds only.

Figure 2 shows the mode shapes and the frequency comparison; Figure 3 shows the position- and removal-dependence of the milling-point receptance that motivates the entire approach: the modal participation $b_f(x_T)$ of the dominant modes varies by design-relevant factors along the pass, exactly the variation that a frozen controller must absorb as uncertainty.

Table 2: Natural frequencies of the clamped plate: present FEM versus the measured values and the theoretical model of the reference [13].

Mode	FEM (Hz)	Measured (Hz)	Error (%)	Ref. model (Hz)	Ref. error (%)
1	519.5	540	3.8	537	0.6
2	1052.1	1068	1.5	1101	3.1
3	2683.5	2787	3.7	2805	0.6
4	3289.7	3351	1.8	3423	2.1
5	4040.3	4122	2.0	4254	3.2

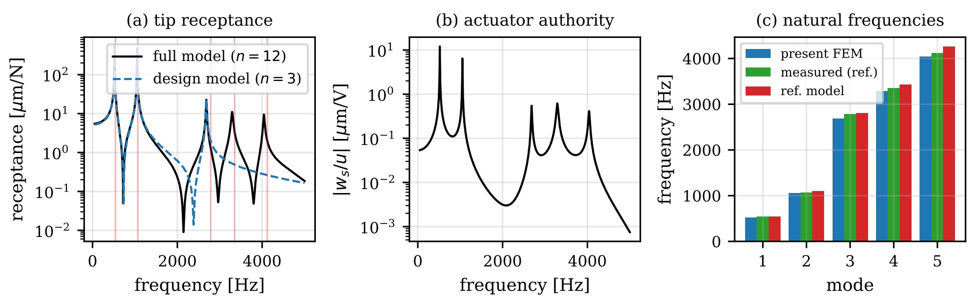

Figure 2: Model validation: mode shapes of the clamped plate and comparison of the finite-element natural frequencies with the measured values of the reference rig [13].

3. Controller synthesis

Figure 4 summarizes the design framework: a grid of frozen-parameter H_∞ syntheses over the scheduling rectangle, re-assembled at a common attenuation level for well-posed interpolation, augmented by a scheduled regenerative-targeted delayed feedback term, and certified by the three deployment checks of Section 3.6.

3.1. Generalized design plant and weights

At frozen θ, the design plant is Eq. (5) with the cutting-stiffness dyad evaluated at the nominal depth $a_p = 0.3$ mm and the state $x = [\eta, \dot{\eta}]$. The exogenous input collects the residual regenerative force (scaled at 1 N; the performance weight does the shaping) entering through $b_f(\theta)$ and the sensor noise n; the control input is the patch voltage u. The performance channels are $z_1 = W_f(s)\, w_{\text{mill}}$ and $z_2 = W_u(s)\, u$; the measurement is $y = y_s + W_n(s)\, n$. The weights are

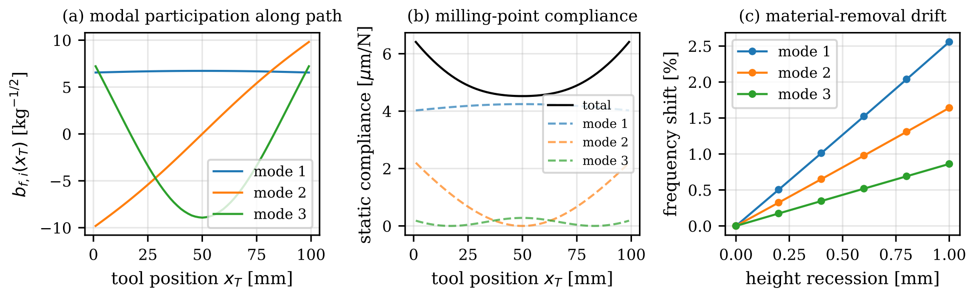

Figure 3: Position- and removal-varying dynamics: milling-point receptance and modal participation $b_f(\theta)$ along the feed path and across material-removal states.

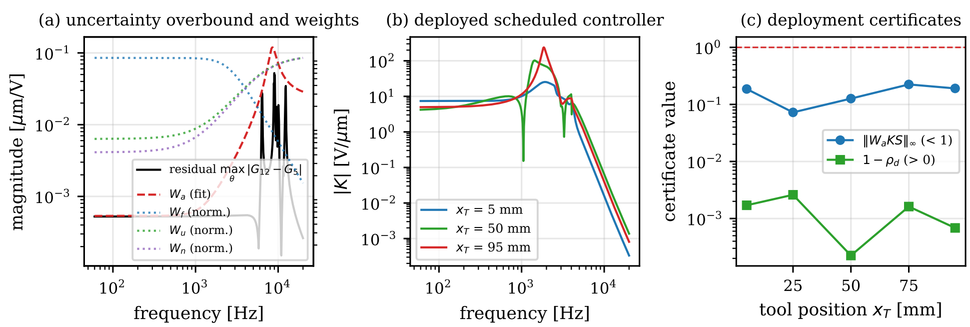

Figure 4: Design framework: grid LPV H_∞ synthesis over $\theta = (x_T, \varrho)$, common-γ re-assembly and gain interpolation, scheduled delayed feedback tuning, and the deployment certificates (small-gain spillover margin, sampled-loop spectral radius, closed-loop semi-discretization lobes).

$$W_f(s) = g_f \, \frac{\omega_f^2}{s^2 + 2\zeta_f \omega_f s + \omega_f^2}, \qquad W_u(s) = g_u \left(\frac{1 + s/\omega_{z,u}}{1 + s/\omega_{p,u}} \right)^2, \qquad W_n(s) = g_n \left(\frac{1 + s/\omega_{z,n}}{1 + s/\omega_{p,n}} \right)^2, \tag{7}$$

with the following values and rationale.

- W_f: second-order low-pass on the milling-point displacement, DC gain $g_f = 2 \times 10^6 \ \mathrm{m}^{-1}$, corner $\omega_f = 2\pi \times 2.5$ kHz, damping $\zeta_f = 0.707$. The corner covers the chatter-relevant band of the retained modes and releases the performance demand above it.
- W_u: lead cascade on the voltage, DC value $g_u = 1/150 \ \mathrm{V}^{-1}$ (the amplifier limit of the reference rig sets the scale), rising by a factor 16 between

$\omega_{z,u} = 2\pi \times 1.5$ kHz and $\omega_{p,u} = 2\pi \times 6$ kHz. The high-frequency penalty discourages controller gain above the retained-mode band.

- W_n: lead cascade on the sensor noise, low-frequency level $g_n = 0.2$ μm (the probe noise floor), rising by a factor 25 between $2\pi \times 1.2$ kHz and $2\pi \times 6$ kHz. Eddy-current probes are noisier at high frequency; more importantly, in the observer-structured central controller it is the rising W_n that bounds the observer gain—and hence the controller gain—in the truncated-mode band. This, together with W_u, is what makes the spillover certificate of Section 3.6 pass.

The weight poles are kept at or below 6 kHz: the weight states become controller states, and every controller pole must remain well below the real-time Nyquist frequency (25 kHz) or the discrete implementation aliases it (this is certified explicitly by the sampled-loop test below).

Truncated-mode spillover is quantified by a fitted additive uncertainty weight. On the voltage-to-sensor channel, the truncation residual $\max_{\theta\in\mathcal{P}} |G_{12}(\mathrm{j}\omega;\theta) - G_5(\mathrm{j}\omega;\theta)|$ between the 12-mode and 5-mode models is computed over the removal states of the grid on a logarithmic frequency grid spanning 50 Hz–20 kHz, and overbounded (with a 2% margin, least-conservative within the candidate set) by the biproper second-order form used in the reference [13]:

$$W_a(s) = r\,\frac{s^2 + 2z_1 w_1 s + w_1^2}{s^2 + 2z_2 w_2 s + w_2^2}, \qquad |W_a(\mathrm{j}\omega)| \geq \max_{\theta\in\mathcal{P}} \big|G_{12}(\mathrm{j}\omega;\theta) - G_5(\mathrm{j}\omega;\theta)\big|. \tag{8}$$

Cutting-coefficient dispersion ($\alpha_4 \in [0.3, 2.9]\,\bar{\alpha}_4$) and modal tolerances ($\pm 10\%$ mass/stiffness, -20% damping) are genuinely uncertain and are verified a posteriori by Monte-Carlo over the closed-loop stability lobes (Section 5.6).

3.2. Actuator roll-off filter and implementation-latency model

Two implementation-fidelity elements are built into the design plant rather than appended after the fact. First, a mandatory fourth-order actuator roll-off filter,

$$F_r(s) = \left(\frac{\omega_c^2}{s^2 + 2\zeta_c\omega_c s + \omega_c^2}\right)^2, \qquad \omega_c = 2\pi \times 4.8 \text{ kHz}, \quad \zeta_c = 0.7, \tag{9}$$

is composed in series with every synthesized controller at deployment; it exploits the 1.55× frequency gap above mode 5 to separate the retained band from the truncated modes. Second, the controller runs on a real-time target at $f_s = 50$ kHz, and the total implementation latency of $1.5/f_s = 30\ \mu$s (half a sample of zero-order-hold reconstruction plus one sample of computation) is represented inside the design plant by its first-order Padé approximant

$$\mathrm{e}^{-sT_\ell} \approx \frac{1 - sT_\ell/2}{1 + sT_\ell/2}, \qquad T_\ell = 30\ \mu\mathrm{s}, \tag{10}$$

so that the synthesized controller carries its sampled-data phase margins by construction. Omitting actuator and filter dynamics from the loop model is known to falsify predicted stability boundaries [45]; here both blocks are inside the synthesis model, the analysis model, and the certificates. The deployed hardware controller is $u = F_r(s)K(s)y_s$; the Padé block is used in continuous-time analysis only, since the physical target realizes the latency itself.

3.3. H_∞ synthesis with γ back-off and controller order reduction

At each grid point θ_k, a standard two-Riccati (DGKF) H_∞ output-feedback synthesis [58] with bisection on the attenuation level γ is performed on the weighted plant. Feasibility at each level requires stabilizing, positive-semidefinite solutions of both game Riccati equations, the spectral-radius coupling condition, a Hurwitz closed-loop interconnection, and a frequency-gridded norm check; the central controller has the familiar observer/state-feedback structure, which is exploited below for scheduling.

Two implementation-driven modifications to the textbook procedure are documented because both were found to be essential.

γ back-off. The returned controller is the central controller re-solved at $1.3\,\gamma_{\mathrm{opt}}$, not at the bisection optimum. At the feasibility edge the game Riccati solutions blow up and the central controller develops parasitic high-gain fast poles; in this design we observed a 69 kHz controller pole carrying nearly 70% of the controller's frequency-response residue—physically meaningless and unimplementable on any real-time target. Backing off to $1.3\,\gamma_{\mathrm{opt}}$ removes the pathology at almost no performance cost: the closed-loop critical depths of cut change by less than 2% for back-off factors between 1.3 and 3.0. All γ-levels reported in this paper are the certified levels of the backed-off controllers.

Order reduction. Residual fast controller dynamics above $0.3\,f_s$ (15 kHz) are removed by ordered real Schur residualization: the slow modes are retained in the leading Schur block and the fast block is replaced by its static (DC) contribution. Every deployment certificate—the spillover margin, the sampled-loop spectral radius, the closed-loop stability lobes, and the time-domain simulations—is evaluated on the *reduced* controller, so the reduction error is covered by the verification rather than assumed away. Across the scheduling grid a common retained order is enforced (11 states for the reported design, plus the roll-off filter) so that the deployed controller state is identical at every scheduling point and survives gain updates.

3.4. Grid scheduling and gain interpolation

The scheduling grid comprises 9 tool positions $\times$ 3 removal states. A two-pass procedure makes the interpolation well posed. Pass 1 runs the γ-bisection independently at every grid point, yielding the locally optimal levels $\gamma(\theta_k)$. Pass 2 re-assembles every central controller at the *common* level $\gamma_c = 1.1\max_k \gamma(\theta_k)$ in the *same* state basis (design-model modal states plus weight states). Because all grid controllers then share the observer structure and the state basis, and because the common γ removes the discontinuous dependence of the central solution on the local optimum, the grid controllers vary smoothly with θ, and piecewise-bilinear interpolation of the realization over (x_T, ϱ) is well conditioned—unlike naive interpolation of independently synthesized controllers, which is ill-posed under state-basis ambiguity [32]. The interpolated controllers at all mid-grid points are validated by closing them on their frozen weighted design plants and checking the spectral abscissa; grid refinement is applied where $\gamma(\theta)$ peaks. It is emphasized, honestly, that this grid procedure does not by itself constitute a stability proof under parameter variation; the formal certificate is provided in Section 4.

3.5. Regenerative-targeted delayed feedback

The regenerative force in Eq. (5) is $\bar{\alpha}_4\, b_f^{\mathsf{T}}[\eta(t) - \eta(t-\tau)]$. An actuation producing the modal force $-\bar{\alpha}_4\, b_f b_f^{\mathsf{T}}[\eta - \eta_\tau]$ would cancel it exactly, but $\Phi^{\mathsf{T}}\boldsymbol{\Theta}$ and $b_f(\theta)$ are not collinear—the actuator is a patch at the clamped corner, the disturbance acts at the moving milling point—so perfect cancellation is impossible and the practical question is how to shape a realizable delayed term. The structure adopted is

$$u_d(t) = k_r(\theta)\left[\hat{w}_{\mathrm{mill}}(t) - \hat{w}_{\mathrm{mill}}(t-\tau)\right], \qquad \hat{w}_{\mathrm{mill}} = b_f(\theta)^{\mathsf{T}}\hat{\eta}, \tag{11}$$

where $\hat{\eta}$ is the state estimate of the H_∞ observer that is already running—no new hardware—and τ is locked to the spindle through the encoder, exactly as in the delayed term of the reference [13] but with three differences: (i) the delayed difference acts on the *reconstructed milling-point* displacement rather than on the fixed sensor point, so it targets the actual regenerative variable; (ii) the gain $k_r(\theta)$ is *scheduled* on the same parameter pair as the H_∞ controller; and (iii) the gain is tuned offline by direct maximization of the closed-loop critical depth of cut,

$$k_r(\theta_k) = \arg \max_{|k| \le k_{\max}} \; a_{\lim}^{\mathrm{CL}}(\theta_k, k), \tag{12}$$

where $a_{\lim}^{\mathrm{CL}}$ is computed by semi-discretization of the complete delayed closed loop (plant, controller, delayed tap, and regenerative delay; Section 4) and $k_{\max}$ encodes the voltage budget. Because $a_{\lim}^{\mathrm{CL}}(k)$ can be multi-modal, a coarse scan followed by local refinement is used rather than a unimodal line search; the tuning is a scalar problem per grid point and is done entirely offline. The total control is $u = u_{H_\infty} + u_d$ with a shared anti-windup saturation at ± 150 V.

3.6. Deployment certificates

Three checks are required of every deployed controller, evaluated on the reduced, filtered controller exactly as implemented.

1. *Small-gain spillover margin.* With $S = (1 - G_5 K)^{-1}$ the sensitivity of the structural loop on the 5-mode model (the interconnection convention throughout this paper is $u = +K y_s$, so an additive plant perturbation sees KS with this sign),

$$\max_{\theta \in \mathcal{P}} \left\| W_a \, K \, S \right\|_\infty < 1, \tag{13}$$

which by the small-gain theorem certifies robust stability against every additive perturbation bounded by the fitted truncation weight W_a of Eq. (8)—that is, against the truncated modes of the 12-mode model over the whole scheduling range.
2. *Exact sampled-loop spectral radius.* The structural plant is discretized exactly (zero-order hold) at the 50 kHz control rate, the deployed controller is discretized exactly, and one full control period of computation delay is appended (the voltage applied during interval k is the value

computed at tick $k-1$, the same timing the real-time target and the simulator implement). The spectral radius of the resulting discrete closed loop must satisfy $\rho < 1$ at every grid point. This certifies the discrete implementation without recourse to the Padé approximation used in synthesis.

3. *Closed-loop stability lobes.* Semi-discretization stability lobes of the full time-periodic delayed closed loop, computed on the 12-mode evaluation model with the deployed controller and delayed term included, along the tool path (Section 4). This is the formal stability guarantee of contribution C3.

3.7. Baselines

Two baselines bracket the proposed controller from below.

Delayed PD (DPD). A delayed PD term on the sensor signal—the single active time-delay control of the reference lineage (Eq. (30) of [13]; see also [41, 42])—implemented through a critically damped second-order low-pass at 8 kHz, with gains tuned by the same closed-loop-lobe maximization as Eq. (12).

Best-frozen H_∞. The strongest *non-scheduled* controller from the same design family: a DGKF controller is synthesized at every grid point with the identical weights, every candidate is evaluated by its worst-case weighted norm $\max_\theta \|T_{zw}(\theta)\|_\infty$ over *all* grid plants (candidates that destabilize any plant are discarded), and the min–max winner is retained. This baseline is deliberately constructed so that any advantage of the scheduled controller over it is attributable to scheduling alone—same model, same weights, same actuator chain, same certificates—and it emulates, from within the same design family, the single-nominal-design practice of the robust paradigm [13, 14].

The fairness protocol is strict: identical weights, identical noise realizations, identical saturation, and the identical evaluation model (12-mode plant, periodic coefficients, loss-of-contact nonlinearity) for all controllers.

4. Stability analysis of the scheduled delayed closed loop

The closed loop formed by Eq. (5), the deployed controller, and the delayed tap of Eq. (11) is a time-periodic delay-differential system: the regenerative delay τ appears in the plant, the same delay appears in the control law, and the coefficient $\alpha_4(t)$ is τ-periodic. Its stability is assessed by first-order semi-discretization [48, 49, 50]: the period is divided into m intervals, the periodic coefficient is averaged per interval, the delayed terms are approximated to first order, and the spectral radius of the monodromy operator over one period decides stability; the critical depth of cut a_{lim} at a given spindle speed is the largest a_p with $\rho < 1$.

A structural property of the loop keeps this tractable despite the augmented controller dynamics: only two *scalars* are delayed—the milling-point deflection $w_{\text{mill}}(t-\tau)$ driving the plant, and the delayed tap $\hat{w}_{\text{mill}}(t-\tau)$ driving the actuation. The discrete state is therefore augmented with scalar histories,

$$z_k = \big[x_k;\ w_k, \ldots, w_{k-m};\ s_k, \ldots, s_{k-m}\big], \tag{14}$$

with $x = [\eta; \dot{\eta}; x_K]$, so the dimension is $(2n + n_K) + 2(m+1)$ instead of the $(m+1)(2n+n_K)$ of a full-state history scheme—the rank-one structure of the cutting dyad, exploited directly. The tooth-passing period is resolved with $m = 48$ intervals throughout—for the gain tuning of Eq. (12) and for every certification lobe reported in Section 5; doubling to $m = 96$ changes the computed spectral radii by less than 0.05%.

The implementation is validated twice before use. Open loop, it reproduces the classical single-degree-of-freedom milling benchmark lobes [49, 1]. Closed loop with the delay in the control path, its critical depth agrees with the exact characteristic-equation limit of the constant-coefficient (turning-type) problem within 0.8%.

The certificate of contribution C3 is then assembled as follows. Closed-loop stability lobes are computed on the 12-mode evaluation model with the deployed scheduled controller frozen at a dense set of tool positions and removal states along the path—start, quarter, half, three-quarter, and end positions at minimum—so that the stability boundary is known *along the tool path*, in the sense of closed-loop stability-lobe practice [44, 45, 46]. The frozen certificate extends to the slowly varying loop by the quantified scheduling-rate bound of Section 2.5: the parameter completes its full range in 20.4 s against millisecond structural periods, four orders of magnitude of separation, and the time-domain simulations of the moving pass (Section 5.4) close the residual gap between the frozen-lobe certificate and the true time-varying loop. It is this semi-discretization certificate—not the grid LPV synthesis of Section 3.4—that constitutes the formal stability guarantee of the proposed controller.

5. Results

5.1. Open-loop validation

Figure 6 shows the open-loop stability lobes of the validated model at the five reference tool positions. At 4900 rpm the open-loop critical depth ranges from 0.229 mm (near the free plate corners, $x_T = 5$ and 95 mm) to 0.255 mm (mid-span), consistent with the reference experiments, in which uncontrolled cutting on this rig loses stability in the 0.1–0.25 mm band depending on position and speed [13]. The position dependence of the open-loop limit (Figure 8, lower curve) is itself modest at this speed—the strong variation appears in the closed loop, where the controller authority depends on the position-varying observability and disturbance geometry.

The open-loop chain admits a sharper consistency test against the published measurements than the modal comparison of Section 2.6 alone. The reference experiments swept five spindle speeds (4300–6700 rpm in 600 rpm steps) and found the uncontrolled stability limit below 0.1 mm at four of them, with a markedly higher limit at 5500 rpm [13]; their force calibration, moreover, treats the effective regenerative coefficient as $1.6\,\bar{\alpha}_4$ nominally, with excursions to $2.9\,\bar{\alpha}_4$. Figure 5 evaluates the present model on exactly their protocol—the *path-minimum* critical depth, since chatter anywhere along the pass terminates it—under the three published coefficient scalings. With the plain average $\bar{\alpha}_4$ the model sits 2–3 times above the measured band, as expected for an uncalibrated linear-edge coefficient; with the reference's own calibrated scalings the model lands on their measurements: at $2.9\,\bar{\alpha}_4$ the path-minimum limit is 0.087/0.079/0.448/0.116/0.097 mm at the five tested speeds—below or at 0.1 mm at four of five, elevated by a factor of four at 5500 rpm, which is precisely the measured pattern including the 5500 rpm anomaly (a lobe peak of the dominant modes that the model reproduces at the correct speed). The stability chain of the model—FEM modal basis, engagement mechanics, regenerative delay, and the semi-discretization machinery that all closed-loop claims rest on—is thereby anchored to the published experimental record at both ends: modal parameters to within 3.8%, and the measured stability pattern once the published coefficient calibration is applied.

5.2. Certificates of the deployed controllers

All three deployment certificates of Section 3.6 pass for every controller the campaign evaluates. The small-gain spillover margin of Eq. (13) reaches

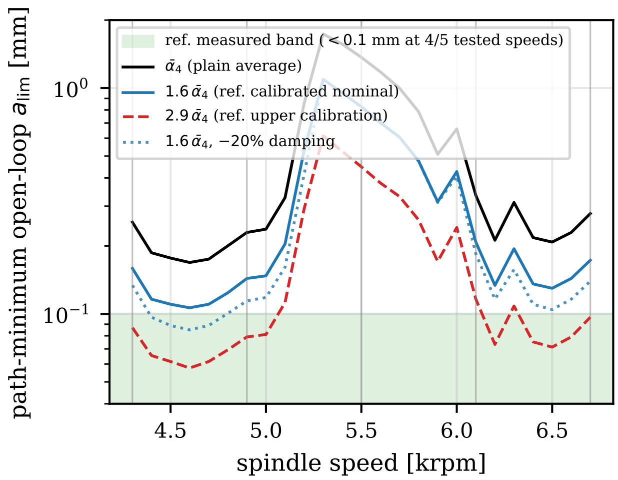

Figure 5: Consistency with the published measurements of the reference rig [13]: path-minimum open-loop critical depth across their tested speed band, for the plain-average regenerative coefficient and for the reference's calibrated nominal ($1.6\,\bar{\alpha}_4$) and upper ($2.9\,\bar{\alpha}_4$) scalings; shaded: their measured range (< 0.1 mm at four of the five tested speeds, elevated at 5500 rpm). Vertical lines mark the tested speeds.

at worst 0.64 across the scheduled grid (at the mid-span position, where the scheduled controller is most aggressive) and 0.22 for the frozen design; the exact sampled-loop spectral radius stays below 0.9998 everywhere (the plant's own lightly damped poles sit at 0.9997–0.9998 per 20 μs period, so the margin is the physical one); interpolated mid-grid controllers close every frozen design plant with spectral abscissa below $-0.7\ \mathrm{s}^{-1}$; and the closed-loop lobes of the next subsection are themselves the third certificate, computed on the full 12-mode evaluation model.

5.3. Closed-loop stability lobes and de-conservatization

Table 3 reports the closed-loop critical depths at 4900 rpm on the full 12-mode evaluation model, and Figures 7 and 8 show the corresponding lobes and the position profile. Three observations carry the headline of the paper.

First, both controllers transform the process: even the frozen design lifts the mid-span limit to or beyond the 5 mm search ceiling, an order of magnitude above open loop. Second, the frozen controller's authority collapses at the end of the pass—0.703 mm at $x_T = 95$ mm, barely three times open loop—precisely where its min-max design point (selected at $x_T = 5$ mm) is

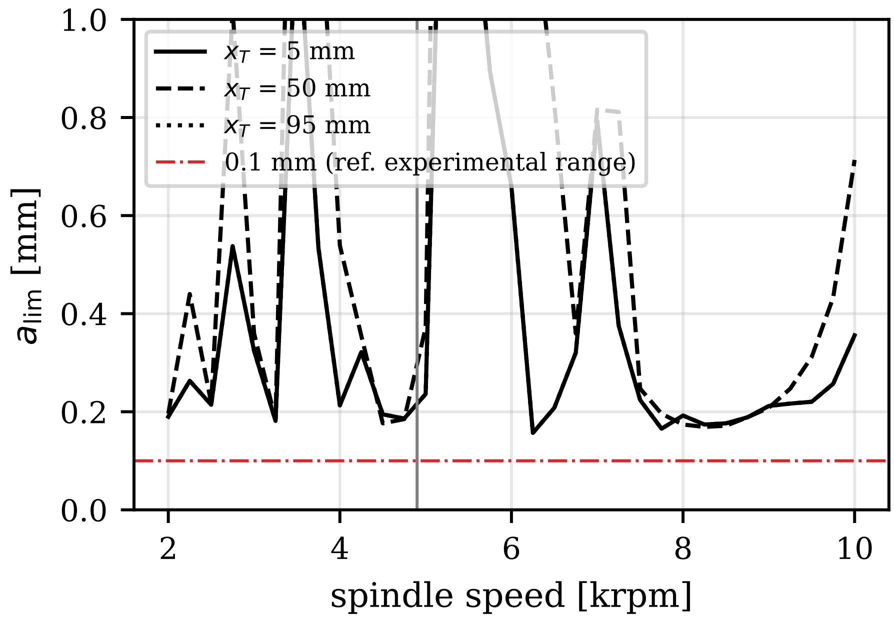

Figure 6: Open-loop stability lobe diagrams of the validated 12-mode model at five tool positions along the feed path.

furthest from the true dynamics. This collapse *is* the conservatism of the non-scheduled paradigm, measured on its most favorable representative. Third, the scheduled controller holds 2.985 mm at its worst position: the worst-position critical depth—the quantity that decides the productivity of a full pass—improves from 0.703 mm to 2.985 mm, a de-conservatization factor of 4.2, and a factor of 13 over the open-loop limit of 0.229 mm. Over the whole 2–10 krpm band the same margin persists: the worst position-and-speed limit is 1.97 mm scheduled against 0.52 mm frozen (Fig. 8b). The delayed-PD baseline manages 0.16–0.98 mm at the reference speed and destabilizes the process at other speeds in the band at every position (its band-minimum is zero; Section 5.5). The γ-levels along the path show the same picture from the synthesis side: the common scheduled level is $\gamma_c = 22.9$ while the best frozen candidate cannot do better than a worst-case $\gamma = 118$ across the same grid (Section 3.4)—a factor of 5.2, quantifying contribution C1 in both currencies, stable depth and attenuation level. A final external anchor: the reference rig's measured stable depth under its (non-scheduled) robust con-

Table 3: Critical axial depth of cut a_{lim} (mm) at 4900 rpm on the full 12-mode evaluation model. Entries of ≥ 5.0 reached the upper limit of the search interval (5.0 mm).

	Tool position x_T (mm)				
Controller	5	25	50	75	95
Open loop	0.229	0.255	0.253	0.255	0.229
Best-frozen H_∞	2.623	4.464	≥ 5.0	4.724	0.703
PS-LPV (scheduled)	2.985	≥ 5.0	≥ 5.0	≥ 5.0	4.695

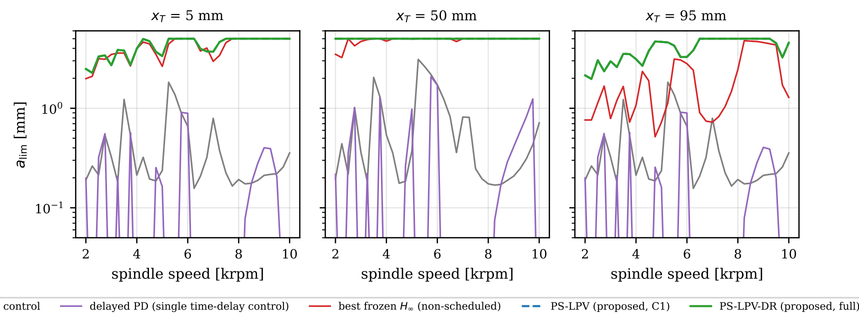

Figure 7: Closed-loop stability lobe diagrams at three tool positions on the 12-mode evaluation model: open loop, delayed PD (tuned at the reference speed; note its destabilized bands at other speeds), best-frozen H_∞, and the scheduled PS-LPV controller (PS-LPV-DR coincides with PS-LPV: the tuned delayed gain is zero).

troller was 0.6–0.8 mm [13]—the same order as the 0.70 mm worst-position limit of the best-frozen design here, as expected for non-scheduled designs of comparable authority class, while the scheduled controller's 2.99 mm shows what the same hardware leaves on the table.

5.4. Time-domain response

Figure 9 shows time-domain simulations of the nonlinear evaluation model (periodic coefficients, loss-of-contact nonlinearity, saturation, measurement noise) at 4900 rpm and $a_p = 0.3$ mm at the start of the pass, $x_T = 5$ mm—which is the min-max design point that the frozen synthesis itself selected. The comparison there is honest about what scheduling does *not* buy: where the frozen design is matched to the local plant, the two controllers coincide within line width (1.22 versus 1.24 μm RMS at 13.8 versus 14.4 V RMS; the uncontrolled cut runs at 6.6 μm in a loss-of-contact limit cycle, and

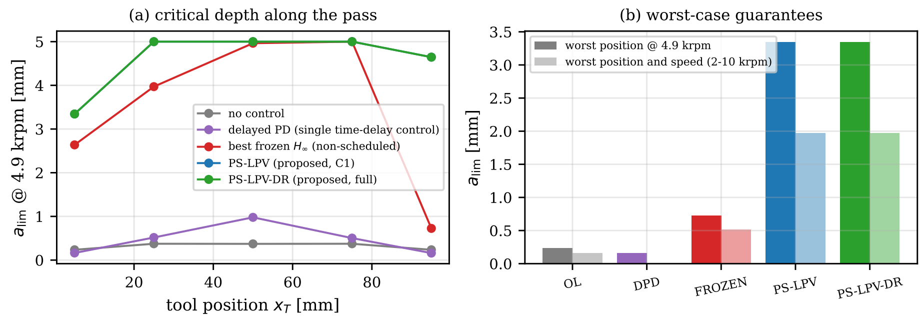

Figure 8: Critical depth of cut at 4900 rpm as a function of tool position: the frozen controller's authority collapses toward the ends of the pass; the scheduled controller sustains it along the entire path.

delayed PD leaves 2.9 μm). Scheduling costs nothing where it is not needed; its value concentrates where the frozen model is wrong. Figure 10 extends this to the full 20.4 s moving pass at $a_p = 1.0$ mm—1.4 times the frozen design's worst-position limit—with the scheduled gains updated continuously along the path: both survive (the loss-of-contact nonlinearity bounds the frozen design's end-of-pass oscillation), with 3.07 μm RMS at 37.0 V for the scheduled controller against 3.40 μm at 29.4 V for the frozen one: the scheduled controller spends its voltage where the plant needs it, the frozen one lets the amplitude grow instead.

The sharpest time-domain separation appears at the condition that the scheduling paradigm predicts: the end of the pass (the frozen design's weakest position) at a depth well above the frozen worst-position limit. Figure 11 shows this condition ($x_T = 95$ mm, $a_p = 2.0$ mm, nearly three times the frozen local limit of 0.703 mm): the open loop and the delayed-PD baseline are violently unstable, the frozen controller survives but labors—15.5 μm RMS at 84.5 V RMS with sustained excursions to the 150 V saturation limit—while the scheduled controller remains comfortable at 6.5 μm RMS and 41.0 V RMS: 42% of the vibration at 49% of the voltage, with saturation margin to spare.

5.5. Contribution of the delayed feedback term

The scheduled delayed term of Eq. (11) is tuned by the lobe-maximizing protocol with a smallest-gain tie preference, so it can only ever help. The certified outcome is unambiguous and, to our knowledge, unreported in the

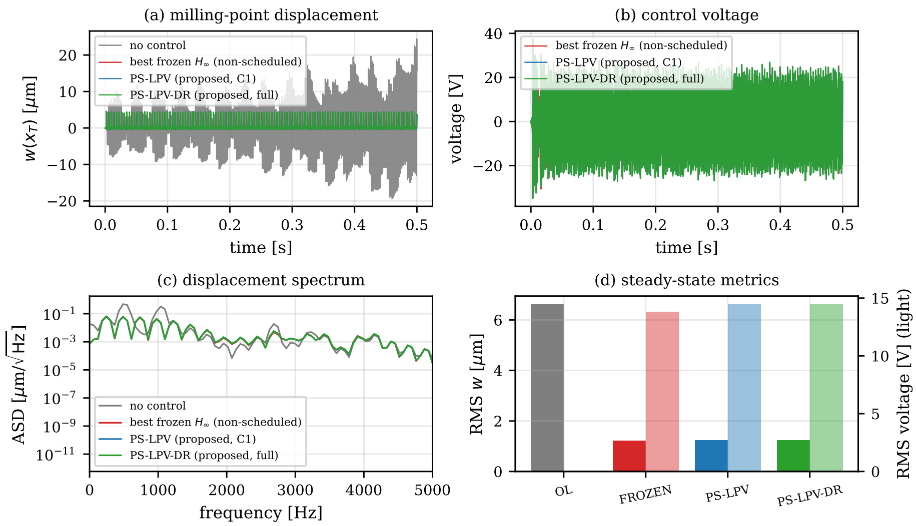

Figure 9: Time-domain response at 4900 rpm, $a_p = 0.3$ mm, at $x_T = 5$ mm—the frozen design's own min-max design point, where the frozen and scheduled controllers coincide by construction: milling-point displacement, control voltage, spectra, and steady-state metrics for all strategies.

delayed-feedback chatter literature: at the reference speed the tuning returns $k_r(\theta) = 0$ at *every* scheduling point—once the scheduled H_∞ loop is active, no exploitable regenerative residue remains that the low-order delayed action of Eq. (11) can profitably cancel, and PS-LPV-DR coincides with PS-LPV in every figure of this section. The finding is not an artifact of the strong scheduled loop: repeating the tuning on top of the frozen controller at its weakest position ($x_T = 95$ mm, $a_{\text{lim}} = 0.703$ mm) yields an increment of only 0.004 mm (+0.5%). Conversely, the same delayed action *alone*—the DPD baseline, tuned by the identical worst-position criterion to $k_p = 0$, $k_d = 60$ V s/m—attains 0.43 mm at the tuning speed but destabilizes the process at other speeds within the 2–10 krpm band at every position (band-minimum $a_{\text{lim}} = 0$, Fig. 7): a fixed low-order delayed gain is speed-fragile, consistent with the reference lineage's own observation that delayed PD alone could not stabilize their cuts [13]. The practical reading for the field: delayed add-ons should be deployed inside a certified loop with a tuning protocol that is free to return zero—and for plants of this class, under a well-designed scheduled H_∞ loop, zero is what it returns. Whether configurations exist (weaker actuators, other sensor placements, other speed regimes) where the

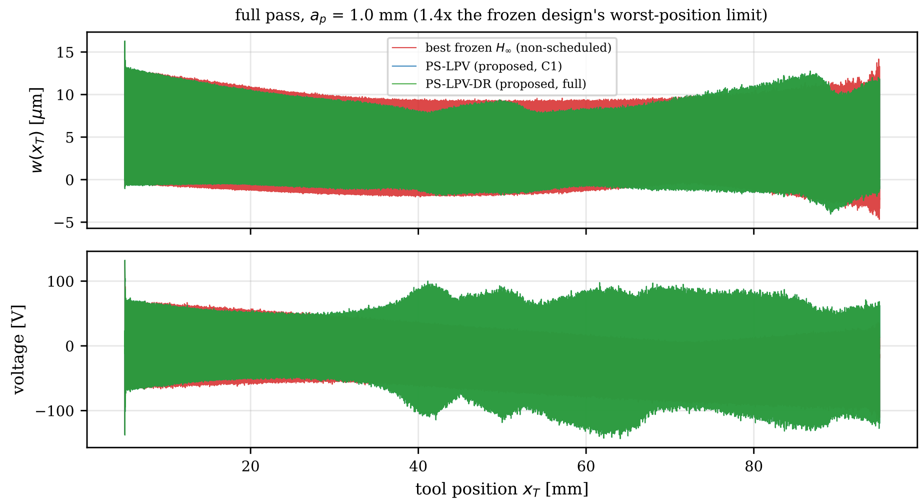

Figure 10: Full moving pass (20.4 s) at $a_p = 1.0$ mm with continuously updated scheduled gains: milling-point displacement and control voltage against tool position, best-frozen H_∞ versus scheduled PS-LPV.

certified optimum is nonzero remains an open, and now precisely posable, question.

5.6. Robustness Monte-Carlo

The scheduling paradigm retains norm-bounded protection only for the genuinely uncertain quantities, so its robustness must be demonstrated, not assumed. Figure 12 shows Monte-Carlo dispersion of the closed-loop critical depth at the frozen design's weakest position under simultaneous dispersion of the regenerative coefficient over $\alpha_4 \in [0.3,\, 2.9]\, \bar{\alpha}_4$—spanning the intermittent engagement's full periodic range—together with $\pm 10\%$ modal mass and stiffness perturbations (mode shapes rescaled coherently with the mass) and up to -20% damping. Over 40 samples the scheduled controller's median critical depth is 2.17 mm against 0.28 mm for the frozen design—a factor of 7.8—and the probability of sustaining the reference depth of 0.3 mm is 68% against 45% (at 1.0 mm: 68% against 25%). The distribution tails are honest in both directions: the most hostile corner of the dispersion box (near $2.9\, \bar{\alpha}_4$ with the damping floor) defeats every controller of this authority class—the lower quartile of both strategies reaches zero—which is the expected physics of a 9.7-fold coefficient sweep, not a scheduling artifact; within the box, scheduling dominates frozen design at every percentile.

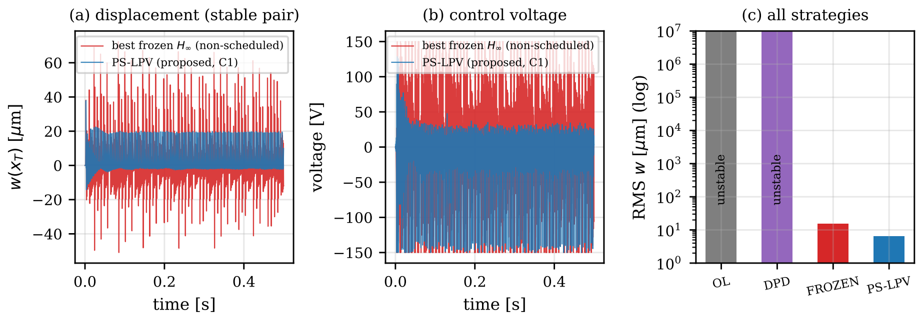

Figure 11: The differentiating condition at the end of the pass ($x_T = 95$ mm, $a_p = 2.0$ mm, 4900 rpm): open loop and delayed PD are unstable; the frozen design survives at high amplitude and voltage; the scheduled controller is comfortable.

5.7. Material-removal scheduling

Over a single 0.1 mm pass the removal-induced modal drift is small and sits comfortably inside the retained uncertainty; the second scheduling variable earns its place over multi-pass sequences. Figure 13 tracks a multi-pass campaign with a total edge recession of 1 mm, over which the natural frequencies of the receding plate drift by 0.9–2.6%. The removal-scheduled controller follows the CAM-predicted state and sustains 2.99–5.0 mm at every position and every removal state. The decisive comparison is the ablation: re-running the *same* scheduled architecture with its gains frozen at the fresh-plate removal state destabilizes the loop outright at mid-pass ($a_{\mathrm{lim}} = 0$ at $x_T = 50$ mm from 0.5 mm recession onward, and at $x_T = 95$ mm at 1 mm)—a point-designed high-authority controller is precisely as fragile to an unmodeled 1–3% modal shift as it is powerful at its design point, which is why the removal axis of the schedule is not optional. The min-max frozen design, robust by construction, survives every removal state but pays permanently at the end of the pass ($a_{\mathrm{lim}} \approx 0.68$–$0.71$ mm). The residual between the CAM-predicted and actual removal state is covered by the retained uncertainty block, which is precisely the division of labor that contribution C1 formalizes: scheduling absorbs the known part, robustness the residual.

6. Saturation-aware certification of the deployed closed loop

The certificates of Sections 3.6–4 govern the *linear* scheduled loop; the piezoelectric amplifier, however, is bounded at $\pm V_{\max} = 150$ V (Table 1),

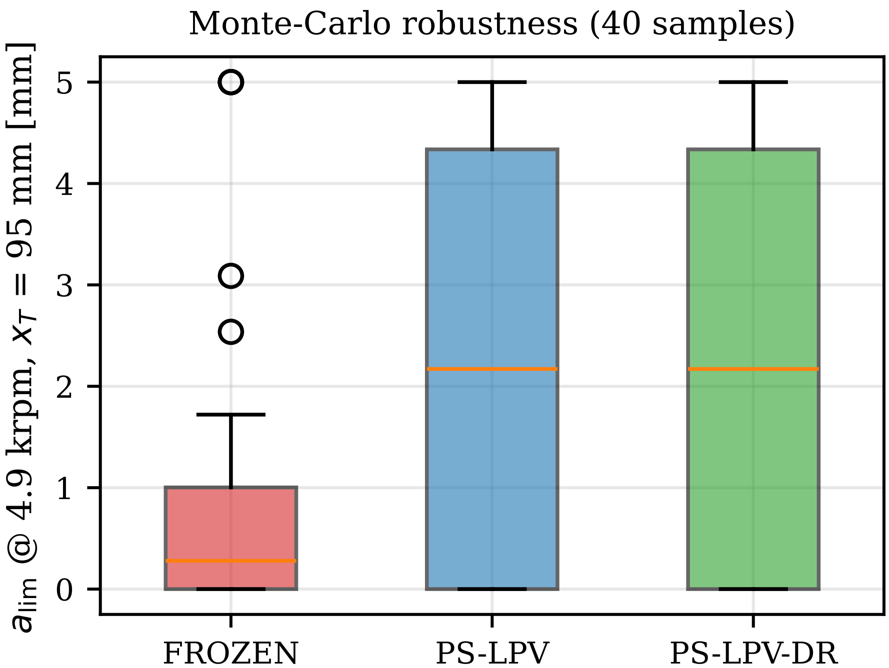

Figure 12: Robustness Monte-Carlo (40 samples, box plots) of the closed-loop critical depth at $x_T = 95$ mm under cutting-coefficient dispersion (0.3–$2.9 \times \bar{\alpha}_4$), $\pm 10\%$ modal mass/stiffness with coherently rescaled mode shapes, and up to -20% damping.

and the evaluation simulations of Section 5 already impose that bound. This section certifies the *saturated* loop: for each operating point it computes the region of surface-defect perturbations over which the deployed controller provably never clips—so the loop stays the linearly certified loop of Section 4—and sweeps that region into a certified permissible depth-of-cut envelope. Under identical machinery the two deployed designs of Section 5 are then re-ranked, with a result the linear analysis cannot see.

6.1. Actuator deadzone and sampled-data period lifting

The amplifier applies $u = \mathrm{sat}(v) = v - \mathrm{dz}(v)$, with v the commanded voltage of the deployed controller and dz the $\pm V_{\max}$ deadzone. The closed loop is lifted per tooth-passing period τ at the real-time control rate $f_c = 50$ kHz: $m = \lfloor \tau f_c \rceil$ ticks of length $\Delta t = \tau / m$ (at 4900 rpm, $m = 204$ and Δt

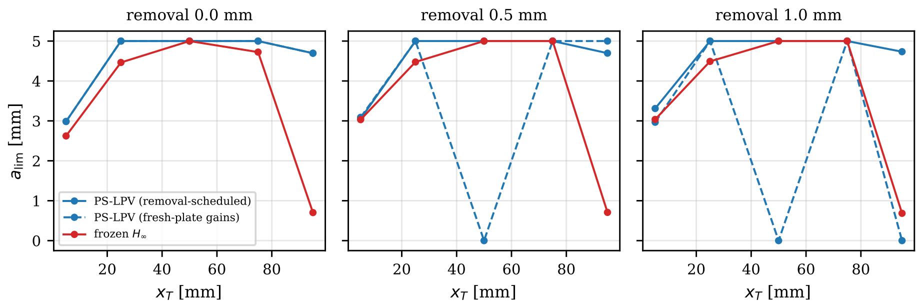

Figure 13: Material-removal study over a multi-pass sequence (total edge recession 1 mm): modal drift of the receding plate and critical depth of the frozen versus removal-scheduled controllers.

deviates from the true control period by 0.04%). The lifted state

$$z = \left[\, x_p;\ x_K;\ u_{\text{pend}};\ w_{1:m} \,\right] \tag{15}$$

collects the plant modal state x_p, the discrete controller state x_K, the pending voltage u_{pend} (the one-period computation delay made explicit, as in the simulator of Section 5), and the ring buffer $w_{1:m}$ of milling-point deflections that realizes the regenerative delay of exactly m ticks. Within each tick the plant advances by its exact zero-order-hold map with the per-tick-averaged coefficient $\alpha_4(t)$ of Eq. (4); the controller output v_k is computed, clipped by the deadzone $\psi_k = \mathrm{dz}(v_k)$, and latched into the pending slot. The homogeneous period map Φ, the affine forced term (the static feed ripple $-\alpha_3 f_t$ of Eq. (4)), the voltage read-out rows R_k, and the deadzone-injection columns are assembled in one pass, so the nonlinearity acts exactly where the physical digital-to-analogue converter clips, with no intra-interval approximation.

6.2. Maximal saturation-free admissible set and the certified surface-step margin

Let z^* be the periodic forced orbit and v_k^* its commanded voltages; z^* is required to be strictly unsaturated, so the phase-resolved headroom $\bar{u}_k = V_{\max} - |v_k^*| > 0$ bounds the shifted deadzone of the variational dynamics about z^*. Any variational trajectory whose commanded voltage never exceeds the local headroom keeps the deadzone inactive, evolves exactly linearly, and decays whenever the sampled Floquet radius $\rho(\Phi) < 1$. The maximal such region is the maximal output admissible set of Gilbert and Tan [53] for the

Table 4: Saturation certificate versus the nonlinear simulator's measured clip onset (PS-LPV, $x_T = 50$ mm, 4900 rpm; both step signs, onset bisected to 6 nm).

a_p	certificate ($\pm$)	model $+h/-h$	simulator $+h/-h$	deviation
0.3 mm	56.8 μm	97.68/56.81 μm	97.57/56.80 μm	0.01–0.11%
1.0 mm	6.9 μm	45.10/6.90 μm	45.07/7.03 μm	0.05–1.8%

period map under the periodic voltage constraint,

$$\mathcal{O}_\infty = \left\{ \delta z : |v_k^* + R_k \Phi^j \delta z| \le V_{\max} \ \ \forall j \ge 0,\ k = 1..m \right\}, \tag{16}$$

invariant by the shift property since v^* is periodic. Its extent along the physical perturbation direction e_h—a surface step of height h left in the stored surface by the previous pass, the milling analogue of the classical initial-condition perturbation—is closed form and certified for both signs:

$$h_{\max} = \min_{j,k} \frac{V_{\max} - |v_k^*|}{\left|R_k \Phi^j e_h\right|}, \tag{17}$$

computed by direct propagation of the unit step in milliseconds, with no semidefinite solver. Periodic generalized-sector matrix-inequality formulations of the same certificate were attempted first and are memory-infeasible at the lifted dimension on commodity hardware; the closed form (17) is not a fallback but tighter in the reported direction. Its declared scope: the step edge crosses at the lift's tick-0 phase; positive-branch values above the feed per tooth certify the contact-retaining model only (a positive step thins the chip toward air cutting); sensor noise is excluded; and the per-tick holds of α_4 and of the delayed surface are covered by the simulator validation next.

6.3. Validation against the nonlinear simulator

The certificate is checked against the full nonlinear time-domain engine of Section 5 (loss-of-contact chip nonlinearity, discrete controller, saturation), with sensor noise and loss of contact disabled to match the contact-retaining scope. The onset of clipping—the smallest injected surface step that saturates the amplifier—is bisected to 6 nm resolution and compared with $h_{\max}$ (Table 4). The certificate equals the binding (negative-step) onset to 0.01–1.8% at these reference points: it is tight, not merely sound.

Table 5: Worst-position depths [mm] at 4900 rpm ($x_T \in \{5, 25, 50, 75, 95\}$ mm). All *certified* worst positions bind at $x_T = 95$ mm; the PS-LPV *linear* worst binds at $x_T = 5$ mm (its linear boundary exceeds the declared 5 mm search cap at the other positions). "Linear" is the sampled-loop Floquet boundary of the same lift.

	linear	@1μm	@2μm	@5μm	@10μm	@20μm	@50μm
best frozen H_∞	0.98	0.99	0.99	0.99	0.91	0.650	0.36
PS-LPV	3.09	1.30	1.16	0.93	0.73	0.513	0.29
ratio	3.16×	**1.32**×	1.18×	0.94×	0.79×	**0.79**×	0.79×

6.4. Certified permissible-depth envelope and the collapse of the linear ranking

Sweeping $h_{\max}$ over the operating grid partitions it, per controller, into four regions: *certified* at a declared tolerance h_{req}; *linear-only* (linearly stable but certificate-refused—the exposure zone where saturation islands can live); *forced-saturated* (the orbit itself clips, refused at any tolerance); and linearly unstable (Fig. 14a). Certified depths are prefix-connected: the reported value is the lower edge of the first refusal, so a feasible pocket above a refused band is never reported as the certified depth.

Table 5 is the central saturation result. The sampled-data lift used here gives a linear worst-position ratio of 3.2× (PS-LPV 3.09 mm at $x_T = 5$ mm versus best-frozen 0.98 mm at $x_T = 95$ mm), close to the continuous-model 4.2× of Section 5.3—the difference is the sampled-loop versus continuous discretization, and every saturation number below is computed within the single sampled-data lift for internal consistency. Once saturation is certified, that 3.2× advantage collapses to at most 1.32×—even at the tightest 1 μm tolerance—and crosses below parity at a $\sim$5 μm tolerance (0.94×), settling at 0.79× from 10 μm upward: the scheduled controller's higher authority amplifies both the forced tooth-passing voltage ripple and the voltage response to surface defects, consuming headroom precisely where the linear analysis promises depth. Forced-saturated zones appear only for the scheduled design (Fig. 14a). This does not reverse the operational verdict: at the hard condition ($x_T = 95$ mm, $a_p = 2$ mm) the scheduled design still dominates in realized vibration and voltage (Section 5.4, Fig. 11); but the certificate adds what the linear picture hides—there the frozen design operates 14.7% clipped against 0.28% for the scheduled design (a ≈50-fold difference), and the frozen loop is in fact linearly *unstable* in the implementation-exact lift ($\rho = 1.41$), its bounded response a purely nonlinear clipped equilibrium.

6.5. *Causal saturation islands*

An operating point is a saturation island when it is linearly stable yet a finite surface-defect perturbation triggers sustained chatter, *and* the same perturbation decays once the amplifier bound is lifted—the causal control that attributes the instability to saturation and nothing else. At 4900 rpm two islands are confirmed, both for the scheduled controller at $x_T = 50$ mm: at $a_p = 1.5$ mm the loop ($\rho = 0.955$) tolerates ± 18.8 μm, but a -37.6 μm step latches into chatter growth at 99.4% clip duty; at $a_p = 2.0$ mm the same occurs at -50 μm (99.5%). In both cases the identical step decays with the bound lifted (Fig. 14d). The frozen design produced no island at the tested points up to ± 500 μm steps. The certified $h_{\max}$ at the island points (1.88 and 0.016 μm) sits 20× and ∼3100× below the island onsets, quantifying the certificate's refuse-to-clip conservatism against the true basin.

6.6. *The saturated regime: recovery margins, a structural limit, and anti-windup*

Beyond the clip boundary the variational loop is $\delta v = \delta v_{\text{free}} + L\,\psi$ with $\psi_k = \mathrm{dz}(v_k^* + \delta v_k)$ in the componentwise sector $[0, 1]$ and L the period-lifted transfer from the deadzone injections to the variational voltages. The discrete circle-criterion level $c = \max_\theta \lambda_{\max}\big(\mathrm{Re}\, L\big)$, evaluated on an eigenvalue-anchored frequency grid, is a sufficient global test: $c < 1$ (with linear stability and an unsaturated orbit) certifies global recovery from arbitrarily deep clipping. Three findings scope the saturated regime honestly. First, a structural impossibility: above the open-loop stability boundary (0.253 mm at $x_T = 50$ mm) no global sector-$[0, 1]$ certificate can exist for any controller—the sector's upper edge is the fully-clipped, i.e. open, loop—so regional certificates above that depth are a necessity, not a conservatism artifact. Second, the measured level is dominated by the controller ($c \approx 1.54$ frozen versus 9.7 scheduled along the $x_T = 50$ mm slice, a sixfold-larger deadzone-loop level for the island-prone high-authority design), though it does not rank every point—the island-free frozen point at $x_T = 95$ mm carries the largest level of all ($c = 13.4$), so worst-case-phase conservatism dominates and the causal experiments of Section 6.5 remain the ground truth. Third, a certified negative result: back-calculation anti-windup enters the lift only through the deadzone-injection columns and provably cannot change any linear certificate; a line search on its gain returns zero improvement of c for both designs at the audited point, so a certified anti-windup benefit requires richer parameterizations or Zames–Falb multipliers [54].

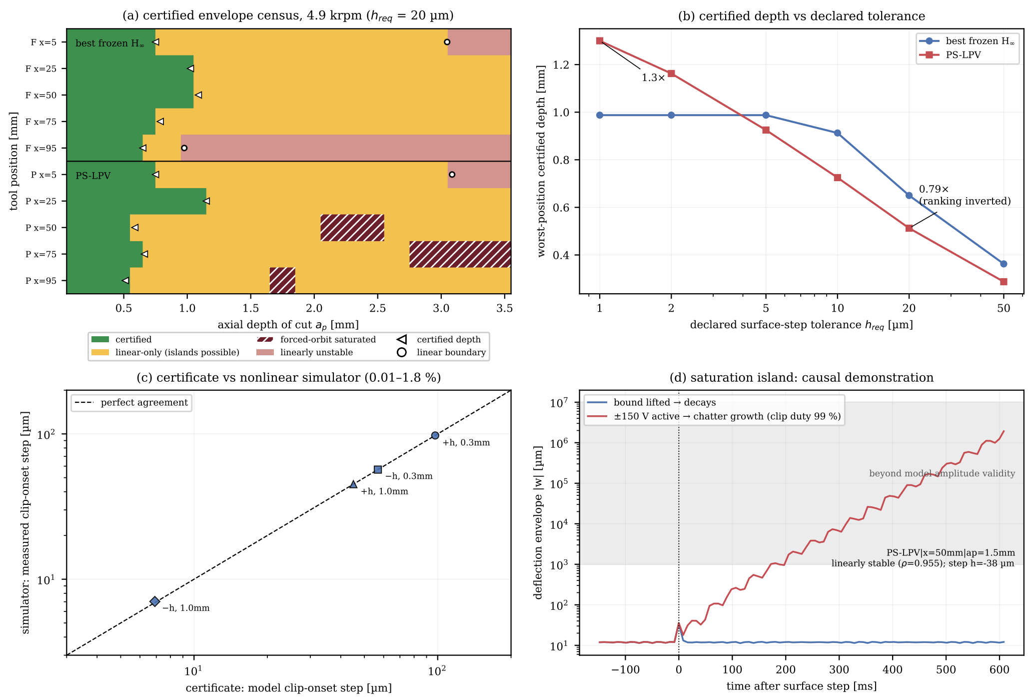

Figure 14: Saturation-aware certification at 4900 rpm. (a) Certified envelope census ($h_{\mathrm{req}} = 20$ μm): certified / linear-only / forced-saturated / unstable zones for both deployed designs. (b) Worst-position certified depth versus declared surface-defect tolerance—the ranking inversion. (c) Certificate versus nonlinear-simulator clip onset, both step signs. (d) Saturation island, causal demonstration: the same -38 μm step decays with the amplifier bound lifted and grows into chatter at 99% clip duty with the ± 150 V bound active.

7. Discussion and limitations

Several limitations bound the claims of this paper, each stated with its mitigation.

Frozen-grid design with a-posteriori certification. The grid LPV synthesis of Section 3.4 carries no global common-Lyapunov/LMI certificate of the kind available for polytopic LPV synthesis [33, 35]; interpolated gains are validated at midpoints and the closed loop is certified by the semi-discretization lobes along the path (Section 4). This route is defensible here because the scheduling rate is three to four orders of magnitude below the plant dynamics—the plate is crossed in 20.4 s against millisecond structural periods—placing the process deep in the slowly-varying regime where frozen-parameter analysis is reliable [31, 32, 34]. An LMI-based synthesis with a bounded-rate certificate, using delayed-LPV machinery [35], is a natural follow-up; the honest position of this paper is that the semi-discretization certificate, not the grid synthesis, is the stability guarantee.

Single-frequency regenerative linearization for design. The design plant uses the time-averaged coefficient $\bar{\alpha}_4$; the full periodic coefficient and the loss-of-contact nonlinearity appear only in the evaluation model. The gap between the two is covered empirically by the evaluation protocol itself (all reported lobes and time-domain results use the full model) and by the Monte-Carlo dispersion of α_4 spanning 0.3–2.9 $\times$ $\bar{\alpha}_4$, which brackets the periodic excursions of the intermittent engagement.

Simulation study anchored to published measurements. The results are numerical, and this paper does not claim experimental validation of the closed loop. What it does claim, and demonstrates, is that every link of the modeling chain is anchored to the published experimental record of the reference rig: modal parameters to within 1.5–3.8% (Section 2.6), and the measured open-loop stability pattern across the tested speed band—including its 5500 rpm anomaly—once the published force calibration is applied (Fig. 5); the closed loop is evaluated with implementation-true timing (sampling, computation delay, saturation, sensor noise) and its non-scheduled representative lands in the same stable-depth class that the rig measured for its robust controller. The closest comparable works are experimental [13, 24, 23]; the hardware path for closing this gap is fully documented—the same plate, patch, amplifier, and sensor already parameterized in Table 1, with the controller on the 50 kHz real-time target for which the latency and sampling certificates were constructed—so that any laboratory equipped with the reference rig's class of hardware can execute the validation from the released design data directly.

Predicted, not measured, removal state. The scheduling variable ϱ is known from the NC program / CAM model, not measured in-process; scheduling-map error (including geometric error of the removal prediction) is covered by the retained additive uncertainty block and the parametric Monte-Carlo. This division—schedule on the known part, remain robust against the residual—is not a workaround but the central design principle of the paper.

Delayed-tap timing in the continuous certificates. The semi-discretization certification enters the delayed feedback tap of Eq. (11) without the implementation latency that the simulator (and the hardware) apply to the total voltage, whereas the H_∞ path carries the full Padé latency model. The approximation is benign at the tuned gain levels—the tap contributes fractions of a volt per micrometre of recirculated displacement, and both delayed strategies (PS-LPV-DR and the delayed-PD baseline) are treated identically within each analysis layer, so the comparisons are internally consistent—but a latency-exact tap certification is a straightforward refinement and is noted for the experimental follow-up.

Nominal-model, refuse-to-clip saturation certificates. The saturation certificates of Section 6 are nominal-model and deterministic, and they refuse rather than exploit clipping: they certify the region over which the loop never saturates, and Section 6.6 shows that above the open-loop boundary this is the only option for a sector-global guarantee. Their conservatism against the true basin is quantified (a factor of 20 to $\sim$3000 at the island points), robustness to cutting-coefficient and modal dispersion is inherited from the Monte-Carlo of Section 5.6, and the sharper Zames–Falb multipliers and a latency-exact tap remain follow-ups. The tolerance-resolved ranking of Table 5 is specific to this benchmark and controller family; its methodological point—that linear margins can misrank controllers under saturation, and that the certified ranking depends on the declared tolerance—is general, and the tolerance-resolved certified envelope is proposed as the object that active chatter-control studies should report alongside stability lobes.

8. Conclusion

This paper replaced the prevailing "known variation as uncertainty" treatment of thin-walled-workpiece milling by "known variation as scheduling, residual variation as uncertainty." A grid-based LPV H_∞ controller, scheduled on the tool position and material-removal state known in real time from the NC program and CAM model, was equipped with a regenerative-targeted delayed feedback term tuned by maximizing the closed-loop critical depth of cut; the scheduled, time-periodic, delayed closed loop—with controller, actuator roll-off, and latency dynamics included—was certified by semi-discretization stability lobes along the tool path, supported by a small-gain spillover margin and an exact sampled-loop spectral-radius test. On a validated model of a benchmark thin-walled plate, scheduling raised the worst-position critical depth of cut from 0.70 mm (best-frozen H_∞ of the same design family) to 2.99 mm—a de-conservatization factor of 4.2 and thirteen times the open-loop limit—held a 3.8-fold margin across the whole 2–10 krpm band, survived a full parametric Monte-Carlo at a 7.8-fold median advantage, and, where the frozen design is weakest, sustained nearly three times that design's local stability limit with 42% of the vibration at half the voltage. The certified delayed-feedback assessment adds a finding of independent value: under an active scheduled H_∞ loop the lobe-maximizing tuning returns a zero delayed gain at every scheduling point—and the same low-order delayed action alone destabilizes the process across the speed band—so delayed add-ons belong inside a certified loop, with a tuning free to say no. Certifying the same deployed loop against its finite amplifier turned actuator saturation from a hidden failure mode into a planning constraint: a solver-free maximal-admissible-set certificate, tight to the measured clip onset within 1.8%, sweeps into a permissible depth-of-cut envelope that reorders the two designs in *guaranteed saturation-free depth*—the scheduled controller's sampled-loop linear advantage (3.2×, the same worst-position comparison that is 4.2× in continuous-model depth) collapsing to 1.32× and inverting to 0.79× at loose tolerance, even though its realized performance advantage persists—predicts and causally confirms saturation islands, and reproduces the published island measurements of a separate rig (Appendix A) on their own parameters. The quantified conclusion is methodological and transfers beyond this rig: in thin-wall milling, the dominant plant variation is known, and a controller that schedules on it recovers the depth of cut and the voltage budget that worst-case designs spend insuring against information they already have.

Appendix A. External validation: reproduction of the published saturation islands

The saturation results of Section 6 rest on the thin-walled plate benchmark alone. This appendix adds an independent external anchor by reproducing measured saturation islands from a *different* rig on its own published parameters, with the same lifting and certificates and no plate-specific tuning.

The island measurements of Ozsoy et al. [52]—a robotic-milling flexure (f_n = 128.1 Hz, ζ = 1.48%, $k = 1.13 \times 10^7$ N/m) under direct velocity feedback (253 V/(m/s)) through an 8.4 Hz proof-mass actuator (3 N/V) with a 27 N saturation threshold, cutting Al 7075-T6 with a 4-flute ∅16 mm 45°-helix cutter at half-immersion down-milling, $f_t = 50$ μm—are re-simulated on the published parameters with the same lifting and certificates (single-mode reduction, untabulated edge coefficients set to zero, 10 kHz feedback rate assumed, and the regenerative sign, absent from the published regeneration-free model, calibrated once against the published uncontrolled boundary; all declared). The calibrated model reproduces the published uncontrolled boundary shape in the island band—a high left flank, model 14–18 mm at 1800–1950 rpm against the published flank rising to $\sim$24 mm at 1950 rpm, collapsing to $\sim$2 mm above 2050 rpm—and then, with no further tuning (Fig. A.1):

- of thirteen actuator-force amplitudes read from the paper's island figure, ten admit a forced-response comparison (the other three lie above the calibrated model's linear boundary where the experiment measured cuts—a declared discrepancy of the single-mode reduction); over the ten, the median model/measured ratio is 1.04, eight cells within $\pm$30% and a worst case of +55% at 1900 rpm / 5 mm;

- the two principal published islands appear as the certificate census's forced-saturated zones—the central island at its 4–8 mm rows (model 1900–1950 rpm; the published extent reaches $\sim$1850–2000 rpm and includes measured saturation at the 10 mm row that the model certifies saturation-free, a declared miss) and the upper island at 18 mm against the published $\sim$20 mm—with the saturation-free band the paper highlights ($\sim$15 mm) appearing in the model at 10–16 mm; the source figure additionally encircles a third band at $\sim$2050–2250 rpm that the paper attributes to loss of contact rather than forced saturation, and that lies along the present model's stability boundary;

- a causal probe at 1900 rpm / 5 mm runs at 44.6% clip duty with mildly degraded RMS and no chatter, consistent with the paper's saturated-but-stable designations in the island interior.

The reproduction also indicates the mechanism: the published islands are forced-response saturation of a mode lying *inside* the tooth-passing band (128.1 Hz; its first harmonic resonates at 1921 rpm, the measured island center)—exactly the forced-saturated zones the present machinery predicts, with regeneration included where the published frequency-domain model omits it.

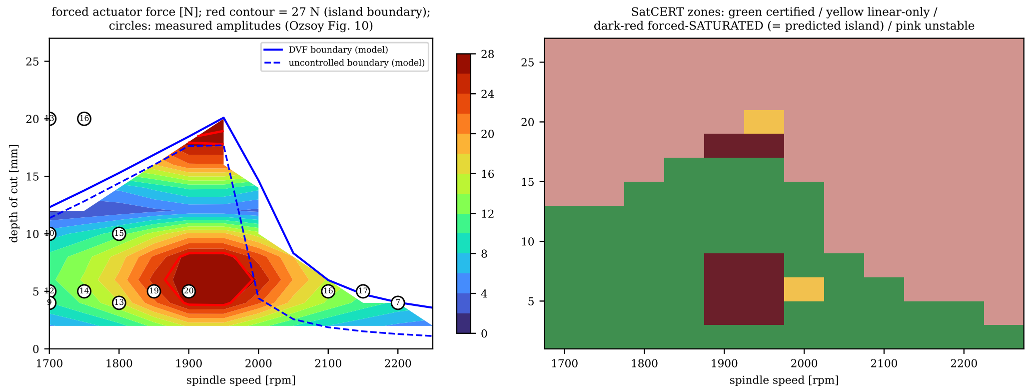

Figure A.1: Reproduction of the published saturation islands of Ozsoy et al. [52] on their own published parameters. Left: forced actuator-force map with the 27 N island contour and the measured amplitudes (circles). Right: certificate zones—the forced-saturated zones are the predicted islands.

CRediT authorship contribution statement

Author One: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing – original draft. **Author Two:** Conceptualization, Supervision, Resources, Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The complete simulation code—finite-element model, controller synthesis, semi-discretization stability analysis, and time-domain engine—together with the scripts generating every figure and table of this paper—together with the digitized measured frequency-response data of the reference rig used in Figures 1 and 5—is available in a public repository at `github.com/samo14s/active-vibration-contr`

References

[1] Y. Altintas, E. Budak, Analytical prediction of stability lobes in milling, CIRP Annals - Manufacturing Technology 44 (1) (1995) 357–362.

[2] Y. Altintas, Manufacturing Automation: Metal Cutting Mechanics, Machine Tool Vibrations, and CNC Design, 2nd Edition, Cambridge University Press, Cambridge, 2012.

[3] Sun, et al., A review of chatter suppression in thin-wall milling: strategies, mechanisms, and applications, International Journal of Extreme Manufacturing (2025).

[4] Yi, et al., Review of milling chatter in aerospace thin-walled structures, Proceedings of the Institution of Mechanical Engineers, Part B: Journal of Engineering Manufacture (2025).

[5] Qin, et al., Chatter stability prediction methods in the machining processes: a review, International Journal of Advanced Manufacturing Technology (2025).

[6] K. Ahmadi, F. Ismail, Stability maps along the toolpath in thin-wall milling, CIRP Journal of Manufacturing Science and Technology (2012).

[7] Y. Yang, et al., A Gaussian process regression-based surrogate model of the varying workpiece dynamics for chatter prediction in milling of thin-walled structures, International Journal of Mechanical System Dynamics (2022).

[8] S. Maslo, et al., Improving dynamic process stability in milling of thin-walled workpieces by optimization of spindle speed based on a linear parameter-varying model, Procedia CIRP 93 (2020).

[9] Y. Zhang, N. D. Sims, Milling workpiece chatter avoidance using piezoelectric active damping: a feasibility study, Smart Materials and Structures 14 (2005) N65–N70.

[10] L. Iorga, H. Baruh, et al., A review of H_∞ robust control of piezoelectric smart structures, Applied Mechanics Reviews 61 (2008).

[11] X. Zhang, C. Zhang, J. Liu, et al., Robust active control based milling chatter suppression with perturbation model via piezoelectric stack actuators, Mechanical Systems and Signal Processing 120 (2019) 808–835.

[12] S. Wang, Q. Song, Z. Liu, Vibration suppression of thin-walled workpiece milling using a time-space varying PD control method via piezoelectric actuator, International Journal of Advanced Manufacturing Technology 105 (2019) 2843–2856.

[13] J. Du, X. Liu, H. Dai, X. Long, Robust combined time delay control for milling chatter suppression of flexible workpieces, International Journal of Mechanical Sciences 274 (2024) 109257.

[14] J. Du, X. Long, Chatter suppression for milling of thin-walled workpieces based on active modal control, Journal of Manufacturing Processes 84 (2022) 1042–1053.

[15] N. J. M. van Dijk, N. van de Wouw, E. J. J. Doppenberg, et al., Robust active chatter control in the high-speed milling process, IEEE Transactions on Control Systems Technology 20 (2012) 901–917.

[16] N. J. M. van Dijk, N. van de Wouw, H. Nijmeijer, Fixed-structure robust controller design for chatter mitigation in high-speed milling, International Journal of Robust and Nonlinear Control 25 (17) (2015) 3495–3514.

[17] E. Mizrachi, S. Basovich, S. Arogeti, Robust time-delayed H_∞ synthesis for active control of chatter in internal turning, International Journal of Machine Tools and Manufacture 158 (2020) 103612.

[18] P. Ruttanatri, M. O. T. Cole, R. Pongvuthithum, H_∞ controller design for chatter suppression in machining based on integrated cutting and flexible structure model, Automatica 130 (2021) 109643.

[19] D. Li, H. Cao, X. Chen, Active control of milling chatter considering the coupling effect of spindle-tool and workpiece systems, Mechanical Systems and Signal Processing 169 (2022) 108769.

[20] A. Dumanli, C. E. Okwudire, Active control of high frequency chatter with machine tool feed drives, CIRP Annals - Manufacturing Technology 70 (1) (2021).

[21] A. Dumanli, C. E. Okwudire, Active chatter mitigation by optimal control of regenerative machining process dynamics, IEEE/ASME Transactions on Mechatronics 27 (2022).

[22] T. Huang, L. Zhu, S. Du, et al., Robust active chatter control in milling processes with variable pitch cutters, ASME Journal of Manufacturing Science and Engineering 140 (2018) 101005.

[23] Robust active control of milling chatter under actuator constraints: a spindle speed mapped framework with experimental validation, Journal of Manufacturing ProcessesIn press (2026).

[24] Z. Brand, et al., An active tool holder and robust LPV control design for practical vibration suppression in internal turning, Control Engineering Practice (2025).

[25] R. Kleinwort, et al., Adaptive active vibration control for machine tools with highly position-dependent dynamics, International Journal of Automation Technology 12 (2018).

[26] C. Wang, X. Zhang, X. Chen, Real time FFT identification based time-varying chatter frequency mitigation in thin-wall workpiece milling, International Journal of Advanced Manufacturing Technology 119 (2022) 7403–7413.

[27] K. Nasiri, et al., Chatter suppression in nonlinear milling of a flexible plate-workpiece with attached piezoelectric actuators: SAC-based controller vs optimized type-2 fuzzy controller, Mechanical Systems and Signal Processing (2025).

[28] M. Hanifzadegan, R. Nagamune, Tracking and structural vibration control of flexible ball-screw drives with dynamic variations, IEEE/ASME Transactions on Mechatronics 20 (1) (2015) 133–142.

[29] X. Huang, et al., Polytopic LPV modeling and gain scheduling H_∞ control of ball screw with position- and load-dependent variable dynamics, Precision Engineering (2024).

[30] C. Onat, M. Sahin, Y. Yaman, Gain-scheduling H_∞ control of a smart beam based on an LPV model, in: 8th ECCOMAS Thematic Conference on Smart Structures and Materials (SMART 2017), Madrid, Spain, 2017.

[31] J. S. Shamma, M. Athans, Guaranteed properties of gain scheduled control for linear parameter-varying plants, Automatica 27 (3) (1991) 559–564.

[32] W. J. Rugh, J. S. Shamma, Research on gain scheduling, Automatica 36 (10) (2000) 1401–1425.

[33] P. Apkarian, P. Gahinet, G. Becker, Self-scheduled H_∞ control of linear parameter-varying systems: a design example, Automatica 31 (9) (1995) 1251–1261.

[34] C. Hoffmann, H. Werner, A survey of linear parameter-varying control applications validated by experiments or high-fidelity simulations, IEEE Transactions on Control Systems Technology 23 (2) (2015) 416–433.

[35] C. de Souza, R. M. Palhares, New gain-scheduling control conditions for time-varying delayed LPV systems, Journal of the Franklin Institute 359 (2) (2022) 719–742.

[36] D. Wang, S. Wu, L. Wan, G. S. Dulikravich, Time-delay LPV system control and its application in chatter suppression of the milling process, Mathematical Problems in Engineering 2015 (2015) 307149. doi:10.1155/2015/307149.

[37] N. Olgac, B. T. Holm-Hansen, A novel active vibration absorption technique: delayed resonator, Journal of Sound and Vibration 176 (1) (1994) 93–104.

[38] N. Olgac, M. Hosek, A new perspective and analysis for regenerative machine tool chatter, International Journal of Machine Tools and Manufacture 38 (1998) 783–798.

[39] T. Vyhlídal, D. Pilbauer, B. Alikoç, W. Michiels, Analysis and design aspects of delayed resonator absorber with position, velocity or acceleration feedback, Journal of Sound and Vibration 459 (2019) 114831.

[40] I. Mancisidor, A. Pena-Sevillano, Z. Dombovari, R. Barcena, J. Munoa, Delayed feedback control for chatter suppression in turning machines, Mechatronics 63 (2019) 102276.

[41] J. Du, X. Liu, X. Long, Time delay feedback control for milling chatter suppression by reducing the regenerative effect, Journal of Materials Processing Technology 309 (2022) 117740.

[42] X. Li, et al., Displacement difference feedback control of chatter in milling processes, International Journal of Advanced Manufacturing Technology (2022).

[43] X. Dong, G. Tu, L. Hu, et al., Suppress chatter in milling of thin-walled parts via fixture with active support, Journal of Vibration and ControlEarly access (2023).

[44] X. Zhang, et al., Discrete time-delay optimal control method for experimental active chatter suppression and its closed-loop stability analysis, ASME Journal of Manufacturing Science and Engineering 141 (2019).

[45] D. Lehotzky, D. Bachrathy, et al., Milling processes with active damping: modeling and stability, Journal of Computational and Nonlinear Dynamics 16 (2021).

[46] J. Monnin, F. Kuster, K. Wegener, Optimal control for chatter mitigation in milling—Part 1: modeling and control design, Control Engineering Practice 24 (2014) 156–166.

[47] J. Monnin, F. Kuster, K. Wegener, Optimal control for chatter mitigation in milling—Part 2: experimental validation, Control Engineering Practice 24 (2014) 167–175.

[48] T. Insperger, G. Stépán, Semi-discretization method for delayed systems, International Journal for Numerical Methods in Engineering 55 (5) (2002) 503–518.

[49] T. Insperger, G. Stépán, Updated semi-discretization method for periodic delay-differential equations with discrete delay, International Journal for Numerical Methods in Engineering 61 (1) (2004) 117–141.

[50] T. Insperger, G. Stépán, Semi-Discretization for Time-Delay Systems: Stability and Engineering Applications, Vol. 178 of Applied Mathematical Sciences, Springer, New York, 2011.

[51] F. Borgioli, D. Hajdu, T. Insperger, G. Stépán, W. Michiels, Pseudospectral method for assessing stability robustness for linear time-periodic delayed dynamical systems, International Journal for Numerical Methods in Engineering 121 (2020).

[52] M. Ozsoy, N. D. Sims, E. Ozturk, Actuator saturation during active vibration control of milling, Mechanical Systems and Signal Processing 224 (2025) 111942. doi:10.1016/j.ymssp.2024.111942.

[53] E. G. Gilbert, K. T. Tan, Linear systems with state and control constraints: the theory and application of maximal output admissible sets, IEEE Transactions on Automatic Control 36 (9) (1991) 1008–1020. doi:10.1109/9.83532.

[54] S. Tarbouriech, G. Garcia, J. M. Gomes da Silva Jr., I. Queinnec, Stability and Stabilization of Linear Systems with Saturating Actuators, Springer, London, 2011.

[55] L. Fu, B. Zhou, G.-R. Duan, Maximized domain of attraction for time-delay systems with actuator saturation, Asian Journal of Control 15, bibliographic details to be verified at submission (2013).

[56] D. Altshuller, Delay-integral-quadratic constraints and stability of time-periodic feedback systems, SIAM Journal on Control and Optimization-Bibliographic details to be verified at submission (2008).

[57] Y. Wu, H. Zhang, T. Huang, G. Zhao, H. Ding, Adaptive chatter control for milling processes with input saturation, International Journal of Robust and Nonlinear Control 26, bibliographic details to be verified at submission (2016).

[58] J. C. Doyle, K. Glover, P. P. Khargonekar, B. A. Francis, State-space solutions to standard H_2 and H_∞ control problems, IEEE Transactions on Automatic Control 34 (8) (1989) 831–847.